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Inference of germinal center evolutionary dynamics via simulation-based deep learning

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This paper presents a computational method to infer from data a key feature of affinity maturation: the relationship between the affinity of B-cell receptors and their fitness. The approach, which is based on a simple population dynamics model but inferred using AI-powered Simulation-Based Inference, is novel and **valuable**. It exploits recently published data on replay experiments of affinity maturation. The method is well argued and presented, and the validation is **compelling**.

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Abstract

B cells and the antibodies they produce are vital to health and survival, motivating research on the details of the mutational and evolutionary processes in the germinal centers (GCs) from which mature B cells arise. It is known that B cells with higher affinity for their cognate antigen (Ag) will, on average, tend to have more offspring. However, the exact form of this relationship between affinity and fecundity, which we call the “affinity-fitness response function”, is not known. Here we use deep learning and simulation-based inference to learn this function from a unique experiment that replays a particular combination of GC conditions many times. All code is freely available at <https://github.com/matsengrp/gcdyn>, while datasets and inference results can be found at <https://doi.org/10.5281/zenodo.15022130>.

Introduction

The germinal center (GC) is the site of affinity maturation of B cell receptors (BCRs), and as such is of central importance to the functioning of the adaptive immune system. Naive B cells, after encountering their cognate antigen, migrate to GCs, where they alternate cycles of affinity-based selection (in the GC’s light zone) and proliferation and mutation (in the dark zone) (Cyster and Allen, 2019; Victora and Nussenzweig, 2022). Over time scales of several weeks (and many generations) this results in an increase in the average affinity of the population of B cells (Berek and Milstein, 1987; Allen et al., 1988). Understanding these processes that occur in the GC is thus of great importance both to achieving a fundamental grasp of the immune system, and to advancing goals such as making a vaccine for difficult-to-neutralize viruses (Burton et al., 2012).

The GC is a Darwinian evolutionary system designed to improve affinity, and as such one might ask: what relationship between affinity and fitness does it use? In order for the GC reaction to improve affinity, this relationship must be increasing. That is, on average, B cells with higher affinity BCRs will reproduce more than those with low affinity BCRs, leading to an overall improvement in affinity. However, the details of the relationship between affinity and fitness are unknown. This is in part because the fitness of a B cell is an emergent property of an intact, functioning germinal center and thus cannot be measured using an in vitro assay.

There have been a variety of suggestions for the form of the affinity-fitness relationship (Batista and Neuberger, 1998 [DOI](#); Kuraoka et al., 2016 [DOI](#); Murugan et al., 2018 [DOI](#)). Resolving these possibilities in practice, however, requires care, even in the ideal situation of single-cell sequencing of individual germinal centers (Tas et al., 2016 [DOI](#); DeWitt et al., 2025 [DOI](#)). One cannot simply examine by hand trees built from observed sequences and gain insight. Indeed, processes such as the “push of the past” and the “pull of the present” (Nee et al., 1994 [DOI](#); Budd and Mann, 2018 [DOI](#)) can mislead interpretation of germinal center sequence data. Instead, an approach is needed that can model extinct lineages that do not appear in the reconstructed tree (DeWitt et al., 2025 [DOI](#)).

We will approach the affinity-fitness relationship via a birth-death model in which the birth rate is a function of affinity; we call this function the “affinity-fitness response function.” We assume a family of such functions, determined by some set of parameters, such that a choice of parameters gives a specific response function. We will use data from the “replay” experiment (DeWitt et al., 2025 [DOI](#)) to fit this function. This experiment used mice whose GCs are seeded entirely by B cells with a single, fixed naive antibody. They were immunized with this naive antibody’s cognate antigen, and the subsequent GC reactions were observed in two ways: by extracting and sequencing individual GCs near a single timepoint, and by pooling multiple GCs at each of a wide range of timepoints for single-cell sequencing. In addition, a deep mutational scan (DMS) on the naive antibody’s sequence was performed for binding to the immunogen. This DMS experiment enables us to make predictions about the affinity of an arbitrary sequence. These affinities can then be input into the affinity-fitness response function; one can compare the observed phylogenetic trees to predictions made by the corresponding birth-death process to learn about the parameters of the response function.

Ideally one would be able to fit the parameters of the affinity-fitness response function via maximum likelihood or Bayesian inference. However, this is difficult. In general, calculating likelihoods for birth-death models is a challenge, requiring solutions of ordinary differential equations (ODEs) that take the probability of unsampled lineages into account. Bakis et al. (2025) [DOI](#) have employed this approach, using the multitype birth-death (MTBD) model of (Kühnert et al., 2016 [DOI](#); Barido-Sottani et al., 2020 [DOI](#)) but adapting it to the case of many trees evaluated in parallel. The types in the MTBD are discretized bins of affinity, and mutation moves lineages between bins.

This approach has the substantial advantage of being rigorously tractable using numerical integration of ODEs and Markov chain Monte Carlo (MCMC); however, the inferential model is misspecified in two ways. First, it requires Markov evolution of the types in the MTBD model. In our case the type is determined by the sequence, and so type evolution is determined by sequence evolution. Because the relationship between sequence and affinity type is non-trivial, and the sequence-based mutation process is complex, evolution of the types is not Markovian. Second, the MTBD model cannot accommodate population size constraints such as those present in real germinal centers. GCs typically reach carrying capacity around 10 days, well before our sampling for these experiments at 15 or 20 days (Schwickert et al., 2007 [DOI](#); Mesin et al., 2020 [DOI](#)). Models with such constraints in fact generally have intractable likelihoods because the evolution of each cell depends on the entire population of cells.

Likelihood-free inference is an alternative approach to fitting complex statistical models. Classical likelihood-free inference uses Approximate Bayesian Computation with summary statistics (Beaumont et al., 2002 [DOI](#)). More recently, techniques using deep neural networks have emerged, especially in population genetics (Sheehan and Song, 2016 [DOI](#); Schrider and Kern, 2018 [DOI](#)). Recent

work has extended this to the inference of phylogenetic model parameters (Voznica et al., 2022 [↗](#); Lajaiti et al., 2023 [↗](#); Landis and Thompson, 2025 [↗](#)) including state-dependent diversification models (Lambert et al., 2023 [↗](#); Thompson et al., 2024 [↗](#)). Can we use these recent methods to learn the parameters of the affinity-fitness function in the germinal center?

In this paper, we learn the relationship between affinity and fitness via simulation-based inference using cell-based forward simulation, neural network inference, and summary statistics matching (Figure 1 [↗](#)). We begin from the compact bijective ladderized vector (CBLV) tree encoding of Voznica et al. (2022) [↗](#), adding affinities for each node from ancestral sequence reconstruction and the sequence-affinity mapping. Our forward model contains too many parameters to be inferred using neural networks alone, so we complement the neural network inference with summary statistic matching for some parameters. We infer response functions on 119 trees from (DeWitt et al., 2025 [↗](#)) and find them to be generally similar across GCs, and roughly tripling from affinity 0 (naive) to 1, and then tripling again from 1 to 2. This means that a cell that mutated to affinity 1 in the early stages of the GC would replicate roughly three times as fast as the cells around it.

Methods

Overview of the replay experiment and data

In order to quantify the mechanics of GC selection we use data from an experimental mouse system that we nickname the “replay” experiment (DeWitt et al., 2025 [↗](#)) in which all naive B cells seeding GCs are identical, carrying the same pre-rearranged IG genes. Thus, the starting sequences for affinity maturation in these GCs have identical affinity and specificity for their cognate antigen. This antigen is chicken IgY, which has been characterized previously (Tas et al., 2016 [↗](#); Jacobsen et al., 2018 [↗](#)). The naive sequence chosen (called “clone 2.1”) was the naive precursor to the dominant sequence in a GC with a very large clonal burst 10 days after immunization (Tas et al., 2016 [↗](#)), suggesting it is a good starting point for affinity maturation. The engineered mice carry the unmutated versions of the Igh and Igk genes rearranged by clone 2.1 in their respective loci. They were immunized with chicken IgY, and the resulting GC reactions were observed in two ways, resulting in two separate data sets. For the first, which we refer to as the “extracted GC data”, 119 individual GCs were extracted and sequenced from 12 such mice at either 15 days (52 GCs) or 20 days (67 GCs) after immunization. In the second, called “bulk data”, multiple GCs from the whole lymph nodes of several mice were pooled together at each of seven time points (from 5 to 70 days after immunization), and then analyzed with droplet-based sequencing. In this paper, we only use the extracted GC data; however we compare our results to others derived from the bulk data using separate inference methods.

We infer trees with IQ-TREE (Minh et al., 2020 [↗](#)) version 1.6.12 using all observed BCR sequences from each GC with the known naive sequence as outgroup, and ancestral sequence reconstruction by empirical Bayes. We use the branch lengths from IQ-TREE directly, without attempting to infer calendar times for the internal nodes.

These BCR trees are annotated with antibody affinities at all nodes as follows. First, we perform ancestral sequence reconstruction for every node of the phylogenetic tree. Because these trees have relatively few mutations, this can be done with low error. We then take affinity values for each single-mutation variant from a DMS experiment (DeWitt et al., 2025 [↗](#)) that used methods similar to Adams et al. (2016) [↗](#). For sequences with more than one mutation, we simply add the effect of each individual mutation (i.e. we assume no epistasis), which was shown to be a reasonable approximation in DeWitt et al. (2025) [↗](#).

Affinity is defined relative to the naive dissociation constant ($K_D^N \simeq 40nM = 4 \times 10^{-8}M$ (Tas et al., 2016 [↗](#))):

$$x = -\log_{10} K_D / K_D^N \quad (1)$$

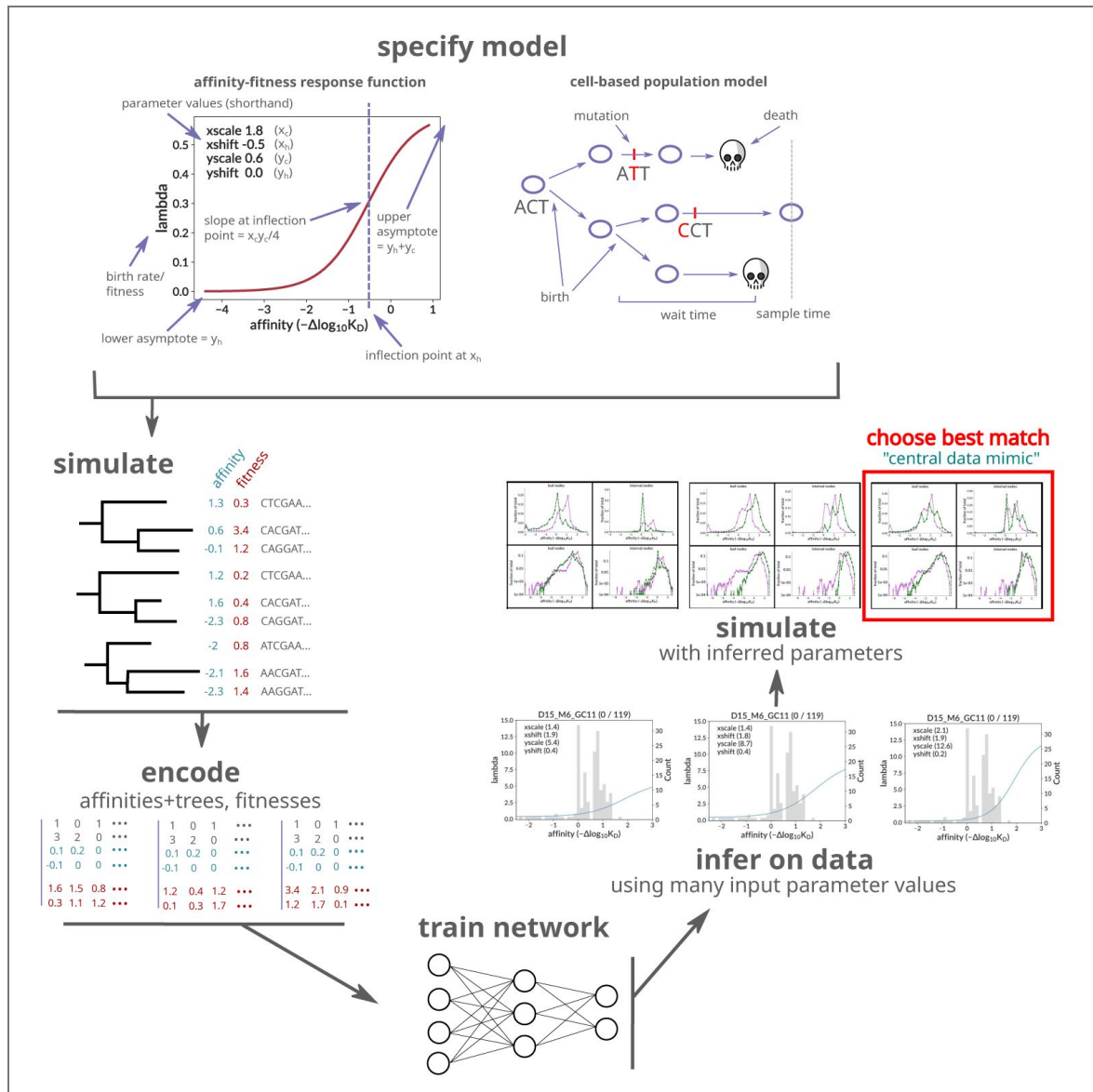


Figure 1. Overview of the workflow for this paper.

After specifying a birth-death model with sigmoid affinity-fitness response function, we simulate many trees and their sequences at each node, with parameters roughly consistent with our real data samples. The simulation is cell-based and implements a carrying-capacity population size limit. The results of the simulation are then encoded and used to train a neural network that infers the sigmoid response parameters on real data. In addition to encoded trees, the network also takes as input the assumed values of several non-sigmoid parameters (carrying capacity, initial population size, and death rate). Inference on real data is performed many times, with many different combinations of non-sigmoid parameter values, and additional "data mimic" simulation is generated using each of the resulting inferred parameter value combinations. The summary statistics of each data mimic sample are compared to real data, with the best match selected as the "central data mimic" sample with final parameter values for both sigmoid (inferred with the neural network) and non-sigmoid (inferred by matching summary statistics) parameters. Plots from elsewhere in the manuscript are rendered in schematic form: those in "infer on data" refer to Figure 4—figure Supplement 1 [\[1\]](#), and those in "simulate with inferred parameters" to Figure 5 [\[2\]](#).

such that, for example, an affinity of +2 means a K_D 0.01 times the naive value. Recall that lower K_D means stronger binding, which corresponds to larger x . Affinity is 0 for the initial unmutated sequence, and ranges from -12.2 to 3.5 in observed sequences, with a mean (median) of -0.3 (0.3). The 119 trees have between 26 and 95 leaves, with a mean (median) of 74 (78) leaves. Observed sequences have between 0 and 19 nucleotide mutations, with a mean (median) of 6.3 (6.0), or around 1%. It is important to note that this is a much lower level than typical BCR repertoires, which average roughly 5-10% nucleotide SHM.

Model

In order to simulate sequences and trees that we use to train the neural network, we employ a birth-death model with a population size constraint. Our model is designed to mimic the replay experiment, with a parameterized affinity-fitness response function mapping each cell's affinity to its fitness. The simulation is initialized with a single root node with the naive sequence from the replay experiment. This initial node then immediately undergoes repeated binary expansion with neither mutation nor time passage until we have the desired number of initial naive sequences.

Each subsequent step in the simulation process consists of choosing one of the three possible events (birth, death, mutation) as follows. We loop over all event types and all nodes, calculating a waiting time for each such node-event type combination. These waiting times are calculated from an event type-specific response function, which relates each node's affinity to the likelihood, or rate, of that event type. For instance the birth response function relates the node's affinity to the rate at which it divides (i.e. at which birth occurs).

The waiting time for each node-event type combination is sampled from the reciprocal of the response function (since response functions are rates, their reciprocal has units of time). For mutation events, the response function is calculated using the SHM model mutabilities summed over each node's sequence. Death events use a two-category response function, with separate (constant) rates for functional and non-functional sequences to reflect rapid purging of nonfunctional cells in the dark zone. The event type-node combination with the smallest waiting time is then selected to occur, and the corresponding cell is removed from the population. If a birth event is selected, two identical children are added to the population. For a death event, no cells are added. For a mutation, the location of the substitution and the new base are chosen according to probabilities from 5-mer sequence contexts in the node's sequence using the model of Cui et al. (2016); a single new cell is created to replace the old.

The process is stopped after a fixed period of time, and a sample of a given size is taken uniformly. If the number of living cells falls to zero, or if there are fewer than 10 living cells at sampling time, the tree is discarded and the simulation is retried.

Inspired by Dessalles et al. (2018), we enforce a carrying capacity on the GC by logistically modulating the birth rate such that the process is critical (average birth and death rates over living cells are equal) when the population is at carrying capacity. We modulate each cell's birth rate by applying a multiplicative factor m :

$$m = \left(\frac{\sum_i \mu_i}{\sum_i \lambda_i} \right)^{N/N_0} \quad (2)$$

which, for carrying capacity N_0 , sums death rates μ_i and birth rates λ_i over the N living cells. For initially small populations, the exponent is near zero and no modulation occurs. As the population increases, however, the exponent goes to one. Thus, as evolution increases affinities and, consequently, birth rates (in the denominator), m becomes small, decreasing each λ_i . Thus initial exponential growth (a supercritical process) gradually slows as it approaches criticality at the specified capacity. If the population fluctuates above carrying capacity, the exponent $N/N_0 > 1$ causes m to decrease rapidly, which in turn decreases the birth rates in its denominator.

We specify the birth response function (relating a node's affinity to its birth rate) as a sigmoid on affinity x (Figure 1—figure Supplement 1):

$$\lambda(x) = \frac{y_c}{1 + e^{-x_c(x-x_h)}} + y_h. \quad (3)$$

The sigmoid's upper asymptote ($y_h + y_c$) represents a hypothesized fitness ceiling above which further affinity increases do not improve fitness. Its lower asymptote (y_h) gives the fitness of very low-affinity nodes (due, for instance, to tonic signaling), which represent cells with either nonfunctional or severely compromised receptors. The x position of the transition between these asymptotes is given by x_h , while x_c determines the transition's steepness. When plotting, we use the following more descriptive names:

- $x_h \rightarrow$ xshift
- $x_c \rightarrow$ xscale
- $y_c \rightarrow$ yscale
- $y_h \rightarrow$ yshift

To construct our training sample, we want to sample x_c , x_h , y_c , and y_h values from throughout biologically plausible ranges. One cannot, however, choose the value of each parameter independently without yielding unrealistic initial birth rates. Besides poorly describing real GCs, such rates would result in many failed simulations. Along with bounds on each of these parameters, we thus also incorporate bounds on the naive (zero-affinity) birth rate, namely

$$\lambda_0 = \frac{y_c}{1 + e^{x_c x_h}}, \quad (4)$$

with lower and upper bounds λ_0^l and λ_0^u . Note that in deriving Eq 4 we have neglected y_h , and instead choose y_h within its own bounds, independently of the other parameters. This amounts to the approximation that very low-affinity nodes have much lower fitness than those with high affinity ($y_h \ll y_c$). We then choose each remaining parameter in an arbitrary, but consistent, order with the following procedure. We first choose a value for x_c from within its bounds $[x_c^l, x_c^u]$ (Table 1), and substitute that value into the λ_0 constraints. This gives us an additional constraint on the next parameter, x_h , which also has its own bounds x_h^l and x_h^u :

$$\frac{\log\left(\frac{y_c^l}{\lambda_0^u} - 1\right)}{x_c} < x_h < \frac{\log\left(\frac{y_c^u}{\lambda_0^l} - 1\right)}{x_c}. \quad (5)$$

Table 1. Simulation parameter values for the sample used for training (left column, in which each GC has different parameters), and the “central data mimic” sample used to evaluate final results (right column, in which all GCs have the same, data-inferred parameters).

Square brackets indicate a range, from which values are chosen either as detailed in the text (for sigmoid parameters) or uniformly at random. The mutability multiplier is an empirical factor that modulates the intensity of mutation to more closely match the speed of evolution to that observed in data.

	training	central data mimic
birth response function	sigmoid	sigmoid
xscale (x_c)	[0.01, 2]	1.6
xshift (x_h)	[-0.5, 3]	2.0
yscale (y_c)	[0.5, 35]	18.2
yshift (y_h)	[0, 0.6]	0.4
carrying capacity	[500, 2000]	500
capacity method	birth	birth
time to sampling	[10, 35]	20
# sampled cells/GC	[50, 130]	[60, 95]

# GCs	50,000	120
naive birth rate	[0.1,15]	1.2
death rate (functional)	[0.05, 0.5]	0.2
death rate (stops)	10	10
initial population	[8-128]	128
mutability multiplier	0.68	0.5

Analogous additional constraints are then also incorporated when subsequently choosing the last parameter y_c , with its own bounds y_c^l and y_c^u :

$$\lambda_0^l (1 + e^{x_c x_h}) < y_c < \lambda_0^u (1 + e^{x_c x_h}). \quad (6)$$

Values for all simulation parameters are listed in [Table 1](#).

Tree encoding

We encode each tree with an approach similar to [Lambert et al. \(2023\)](#) and [Thompson et al. \(2024\)](#), most closely following the compact bijective ladderized vector (CBLV) approach from [Voznica et al. \(2022\)](#). The CBLV method first ladderizes the tree by rotating each subtree such that, roughly speaking, longer branches end up toward the left. This does not modify the tree, but rather allows iteration over nodes in a defined, repeatable way, called inorder iteration. To generate the matrix, we traverse the ladderized tree inorder, calculating a distance to associate with each node. For internal nodes, this is the distance to root, whereas for leaf nodes it is the distance to the most-recently-visited internal node ([Voznica et al., 2022](#), Fig. 2). Distances corresponding to leaf nodes are arranged in the first row of the matrix, while those from internal nodes form the second row.

We further enrich the basic CBLV encoding with information on the state of the evolving entity. In contrast to [Lambert et al. \(2023\)](#) and [Thompson et al. \(2024\)](#), however, who labeled only tips with discrete types, we label all nodes with a continuous variable. The variable we are interested in is affinity, which we incorporate into the encoded matrix as two additional rows. Since the original matrix's two rows represent leaf and internal node heights, each entry in our additional two rows represents the corresponding node's affinity value ([Figure 1](#)).

In addition to repeatability, we want the node ordering to guarantee a unique location for each node in the resulting matrix. In the original method ([Voznica et al., 2022](#)), however, this correspondence (and thus the encoding) is only unique if all nodes have unique times. For generality, we would like to allow the use of ultrametric trees, with all cells sampled at the same time. This has the potential to confuse the neural network, since it destroys the one-to-one correspondence between trees and encodings. We thus add several tiebreakers to the ladderization scheme that establishes the order in which we iterate over nodes. After first sorting nodes by their time, we then sort any ties by the time of their immediate ancestral node, and then further sort any remaining ties by their affinity. This could, in principle, still result in ties. In practice, however, we have set our code to raise exceptions if ties are encountered, and this has not yet occurred.

As for real data, we infer trees on simulation with IQ-TREE ([Minh et al., 2020](#)). By taking only sampled sequences and passing them through the same inference method, we treat data and simulation as similarly as possible, which facilitates accurate neural network training and summary statistic comparison. An alternative would be to compare time trees inferred on data, for instance with BEAST ([Suchard et al., 2018](#)), to simulation truth trees. Although this is commonly done ([Voznica et al., 2022](#)) we reasoned that inference would be more accurate with simulation and data samples on the same footing. Furthermore, our implementation of initial population size

creates large numbers of unmutated ancestral sequences near the root of the simulation truth tree, which would have no observable counterpart in data, where we can only infer the presence of common ancestor nodes.

As in Voznica et al. (2022) [\[14\]](#), we also scale trees to mean unit depth before encoding in order to facilitate comparison across trees of different depths.

Neural network architecture, implementation, and training

Our network is detailed in Table 2 [\[14\]](#), but in brief, after input of encoded trees we begin with several convolutional layers interspersed with pooling layers. There follows a series of dense layers with decreasing complexity until we arrive at the final number of outputs. These dense layers take as input, in addition to the output of previous layers, fixed values for several “non-sigmoid” parameters (i.e. parameters not pertaining to the sigmoid birth response whose parameters we directly infer with the network). The input matrices are padded with zeros to a constant, uniform size (here set to 200), to allow the use of trees with different numbers of nodes. We use the exponential linear unit (ELU) activation function for all layers. Our main “sigmoid” network predicts the four sigmoid parameters. However, as a cross check we also train a model that predicts the actual response function value in discrete bins of affinity, which we call the “per-bin” model. The per-bin model introduces some level of independence from the sigmoid model, however because it is still trained on sigmoid simulation, it has only a limited ability to infer shapes that differ significantly from a sigmoid.

Table 2. Neural network architecture.

order	type	filters	kernel size	pool size	stride length	N units	N params
1	1D convolutional	25	4			426	
2	1D convolutional	25	4			2,525	
3	max pooling			2	2		
4	1D convolutional	40	4			4,040	
5	global avg. pooling						
6	dense					48	1,968
7	dense					32	1,568
8	dense					16	528
9	dense					8	136
10	dense					3	27

In making our training samples we endeavor to cover the entire space of plausible parameter values, which includes choosing bounds on each of the three sigmoid parameters (Table 1 [\[14\]](#)). We find that leaving the inferred parameters entirely unconstrained tends to confuse the neural network, so we impose a “clip function” with bounds on each parameter during network training. For decent performance, these clipping bounds must be somewhat wider than the simulation bounds; but, heuristically, not overly wide (Table 3 [\[14\]](#)).

Table 3. Clip function bounds for neural network training.

parameter	min value	max value
xscale (x_c)	0.001	3.5
xshift (x_h)	-1.5	5
yscale (y_c)	0.1	65
yshift (y_h)	0	10

We scale all input variables (branch lengths, affinities, and non-sigmoid parameters) individually to mean 0 and variance 1 over all trees. We experimented with similarly scaling output parameters, but this did not improve performance.

We use an Adam optimizer with a learning rate of 0.01, an exponential moving average (EMA) momentum of 0.99, and a batch size of 32, values which were arrived at by performing hyperparameter optimization to minimize loss values. We do not use dropout regularization, since in our tests it resulted in worse performance than using a validation sample to measure overfitting. When training, we use 20% of each sample for testing, and also reserve 10% of the remaining 80% as a validation sample. We train for 35 epochs, which for both sigmoid and per-bin networks was the point at which validation sample performance began to decrease. While during development we trained on a large number of simulation samples with different sizes and parameter ranges, the results presented in this paper center on a sample of 50,000 trees with the ranges specified in Table 1. We also used samples as large as 100,000 trees, and with substantially more complex networks, but neither change resulted in significant performance improvements. Also note that these optimization and testing steps were performed only on simulation; we ran data inference only once our methods were essentially finalized.

All code for this paper can be found at <https://github.com/matsengrp/gcdyn>, and all inputs and outputs, along with instructions for running, can be found at <https://doi.org/10.5281/zenodo.15022130>.

Curve difference loss function

For a loss function we use a scaled version of the L^1 distance between the true and inferred functions (Figure 1—figure Supplement 2). This loss, which we refer to as a “curve difference”, is calculated by dividing the area between the curves by the area under the true curve within domain, i.e. affinity bounds, $[-2.5, 3]$. This domain was chosen because below -2.5 most curves are very similar, while above 3 they diverge dramatically in a region in which we have very few measured affinity values. The per-bin model uses the same loss function, but with a coarser discretization (one value for each bin).

An alternate approach would have been to use mean squared error on the inferred parameters, namely the squared difference between inferred and true values over all predictions. In our case, however, this is a poor choice because we care about the sigmoid’s shape, but not about the underlying sigmoid parameter values, which have little individual biological meaning. The curve loss and the mean squared loss on inferred parameters can be quite different: significant simultaneous changes to two parameters can counteract each other, resulting in very similar curve shapes (figure Supplement 3).

Non-sigmoid parameter inference using summary statistics

In addition to the four sigmoid parameters, which we infer directly, there are other parameters in Table 1 about which we have incomplete information. The carrying capacity method and the choice of sigmoid for the response function represent fundamental model assumptions. We also fix the death rate for nonfunctional (stop) sequences, which would be very difficult to infer with the present experiment. For others, we know precise values from the replay experiment for each

GC (time to sampling, # sampled cells/GC), but use a somewhat wider range for the sake of generalizability. The mutability multiplier is a heuristic factor used to match the SHM distributions to data. The naive birth rate is determined by the sigmoid parameters, but has its own range in order to facilitate efficient simulation.

For two of the three remaining parameters (carrying capacity and initial population), we can ostensibly choose values based on the replay experiment. These values carry significant uncertainty, however, partly due to inherent experimental uncertainty, but also because they may represent different biological quantities to those in simulation. For instance, an experimental measurement of the number of B cells in a germinal center might appear to correspond closely to simulation carrying capacity. However if germinal centers are not well mixed, such that competition occurs only among nearby cells, the “effective” carrying capacity that each cell experiences could be much smaller.

Fortunately, in addition to the neural network inference of sigmoid parameters, we have another source of information that we can use to infer non-sigmoid parameters: summary statistic distributions. We can use the matching of these distributions to effectively fit values for these additional unknown parameters. We also include the final parameter, the functional death rate, in these non-sigmoid inferred parameters, although it is unconstrained by the replay experiment, and it is unclear whether it is uniquely identifiable.

We adopt a two-step procedure to infer sigmoid and non-sigmoid parameters, inspired by the concept of conditional generation. Recall that our network takes non-sigmoid parameters as input, i.e. it infers sigmoid values contingent on a set of non-sigmoid values. We thus first infer sigmoid parameters on data for many different values of the non-sigmoid parameters. We then take each resulting set of (inferred) sigmoid and (supplied) non-sigmoid values, generate a new simulation sample with each, and compare their summary statistics to data. These simulation samples, like data but unlike the training sample, consist of 119 GCs with identical parameters, and are referred to as “data mimic” samples. The sample whose summary statistics most closely match data corresponds to the best inferred values for both sigmoid and non-sigmoid parameters, and is called the “central data mimic” sample. While we would ideally include all non-sigmoid parameters in this two-step procedure, the combinatorics of scanning over so many different variables is prohibitive. We thus only use three non-sigmoid parameters: carrying capacity, initial population, and death rate. These were chosen because their values likely include substantial uncertainty, and because they exert significant control over summary statistic distributions. This was implemented as a threedimensional scan over four carrying capacity values (500, 750, 1000, 2000), three initial population values (8, 32, 128), and four death rates (0.05, 0.1, 0.2, 0.4).

Since the summary statistic distributions are affected by all simulation parameters, the best solution would likely be to infer all uncertain parameters with the neural network. This, however, would require passing much more information into a vastly more complicated network (the current tree encoding, for instance, has no information on sequence abundance) that would have to be trained on simulations covering additional dimensions. We thus deemed this approach infeasible for the current paper.

Central (medoid) curve prediction

While our network predicts each sigmoid parameter individually, it is generally only useful to view the resulting curve as a whole. This is because many different sets of sigmoid parameter values can give quite similar shapes, since a change in one parameter can to some extent be compensated by modifying others (Figure 1—figure Supplement 3 [↗](#)). In order to present the results of all GCs together, we thus must compare predictions using curves, rather than individual parameters. We want to choose both a central, representative curve and associated uncertainty bounds that, for each prediction, incorporates all four predicted parameters at once. We do this by calculating the medoid curve among predictions, with distance defined as our curve difference loss function. We thus calculate, for the predicted curve for each GC, the sum of squared differences to all other predicted curves in the affinity bounds $[-2.5, 3]$. We then choose the curve with the smallest such sum as the medoid curve.

Results

Deep learning effectively infers parameters from simulated data

After building and training our neural network, we evaluate its performance on subsets of the training sample. While this evaluation provides an important baseline and sanity check, it is important to note that the training sample differs dramatically from real data (and the “data mimic” simulation sample that mimics real data). While real data consists of 119 GCs with identical parameters and thus response functions, we need the GCs in our training sample to span the space of all plausible parameter values. This means that while we must evaluate performance on individual GCs in the training and testing samples, in real data (and data mimic simulation) we combine results from 119 curves into a central (medoid) curve. Inference on the training sample will thus appear vastly noisier than on real data and data mimic simulation, and also cannot be plotted with all true and inferred curves together.

We show values of the curve difference loss function on the training sample (split into training, validation, and test subsets) for both sigmoid and per-bin models (Figure 2 [↗](#) and Figure 3 [↗](#)). This shows that for both models, the mean area difference on a single GC between true and inferred curves is around 70% of the full plot area. We find that the sigmoid and per-bin models have similar overall performance. We also show true and inferred response functions on four representative GCs with loss values spanning 0.17 to 2.18 (figure Supplement 1 [↗](#)).

Inference on real data

We then apply our neural network to real data. We show the inferred sigmoid response curves on all 119 GCs for both the sigmoid and per-bin models (Figure 4 [↗](#)), as well as several representative individual curve predictions (Figure 4—figure Supplement 1 [↗](#) and Figure 4—figure Supplement 2 [↗](#)). As detailed in the Methods, we run data inference many times, for many different values of three non-sigmoid parameters, generate simulation from each combination, and compare the resulting summary statistics to real data. In Figure 4 [↗](#) we display inference corresponding to the best such parameter values, i.e. the values yielding simulation whose summary statistics best matched data: carrying capacity 500, initial population 128, and death rate 0.2. The summary statistics for the resulting central data mimic sample are shown in Figure 5 [↗](#) and Figure 5—figure Supplement 1 [↗](#).

Finally, we also show curve inference results on the central data mimic sample (Figure 6 [↗](#)). As expected, the curve difference loss values are vastly smaller than for the single-GC results on the training sample (where each GC had a different curve). Here, the medoid sigmoid curve has a loss of 0.09 (i.e. the true and inferred curves differ by 9% of the plot’s area), compared to the mean single-GC loss of 0.7 from the training sample. Also importantly, the variance of inferred curves (blue shading) is of similar magnitude to that in data.

Simulation mimics GC dynamics

In addition to letting us infer non-sigmoid parameter values, the summary statistics distributions also establish how well our simulation mimics real BCR sequencing data. Since for the purposes of this paper we are interested in replay-style experiments, we seek to mimic these particular samples. The summary statistics for the central data mimic sample (Figure 5 [↗](#)) match data reasonably closely. The training sample, on the other hand, matches much less well because it is designed to span all plausible unknown parameter values to ensure that the neural network encounters any parameter combination that it is likely to see in data (Figure 5—figure Supplement 2 [↗](#)). The less-likely parameter combinations, of course, tend to yield poorly-matching summary statistics.

Figure 2. Training and testing results for the sigmoid model on simulation.

We show curve difference loss distributions on several subsets of the training sample (where each GC has different parameter values): training and validation (left) and testing (right). For computational efficiency when plotting, the curve difference distributions display only the first 1000 values. See Figure 3 for per-bin model.

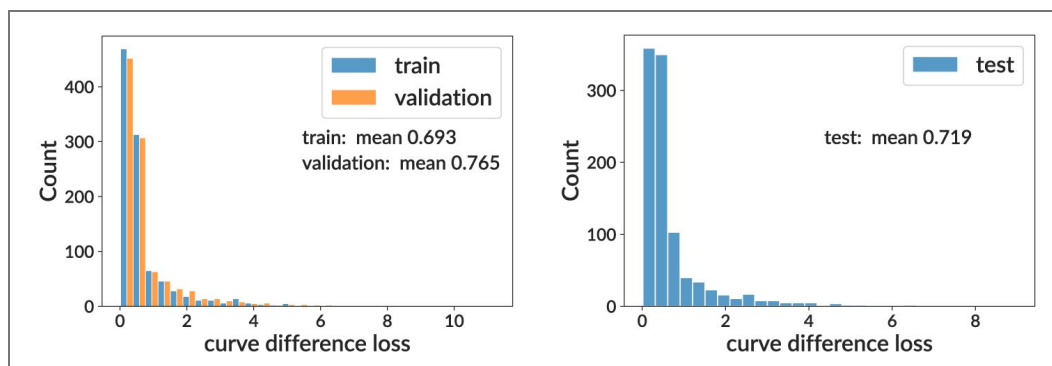


Figure 3. Training and testing results for the per-bin model on simulation.

See Figure 2 for details.

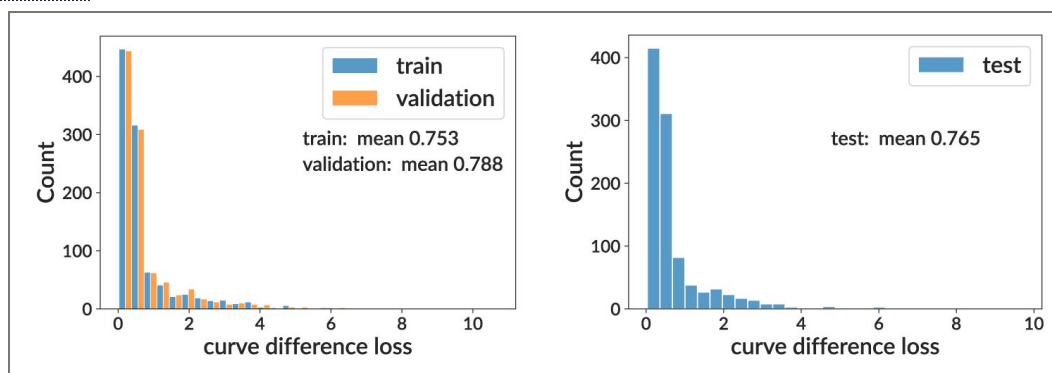
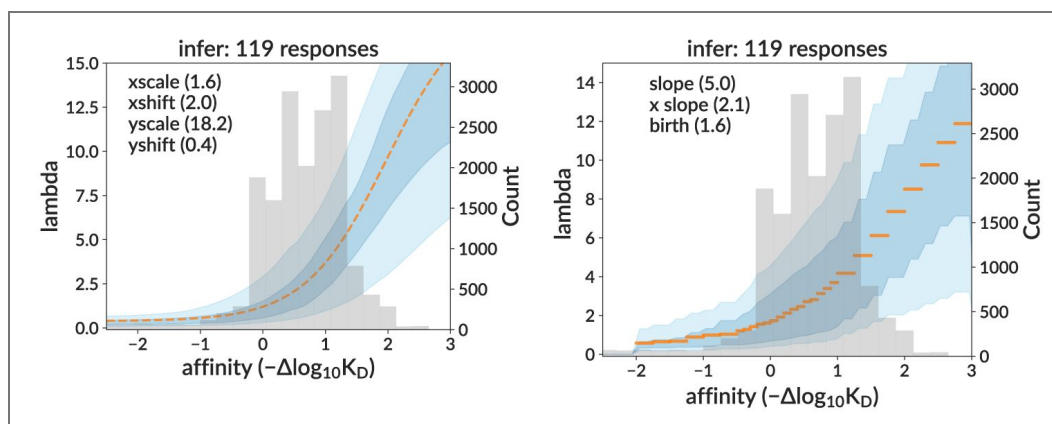


Figure 4. Inferred response functions on real data

for sigmoid (left) and per-bin (right) models, corresponding to the non-sigmoid parameter values yielding simulation with the best-matching summary statistics. The medoid curve is shown in orange with 68% and 95% confidence intervals in blue, with observed affinity values in grey.



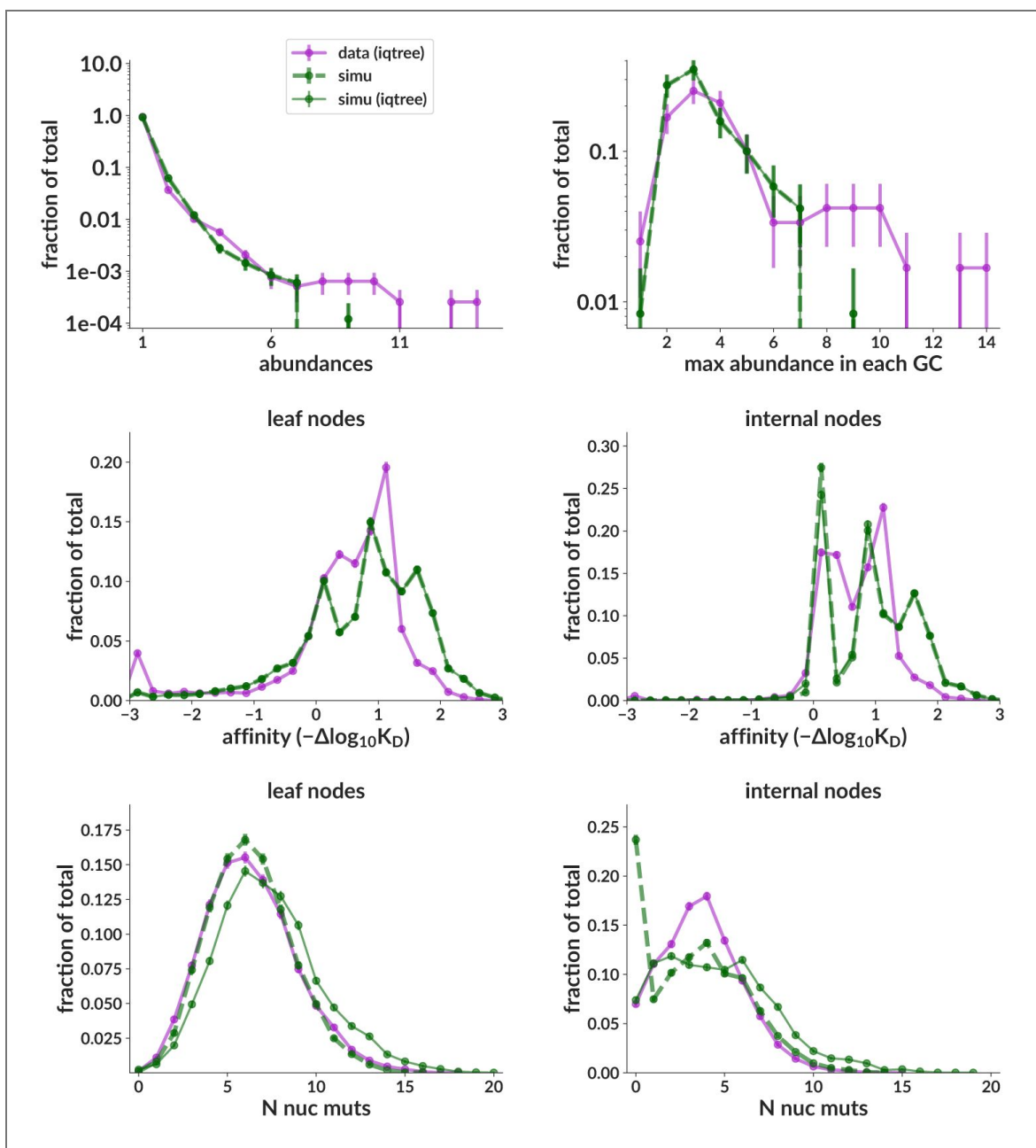


Figure 5. Summary statistics on data vs simulation

for the central data mimic simulation sample that most closely mimics inferred data parameters. Simulation truth (dashed green) is unobservable and shown only for completeness; the important comparison is between purple and green solid lines, where both data and simulation have been run through IQ-TREE.

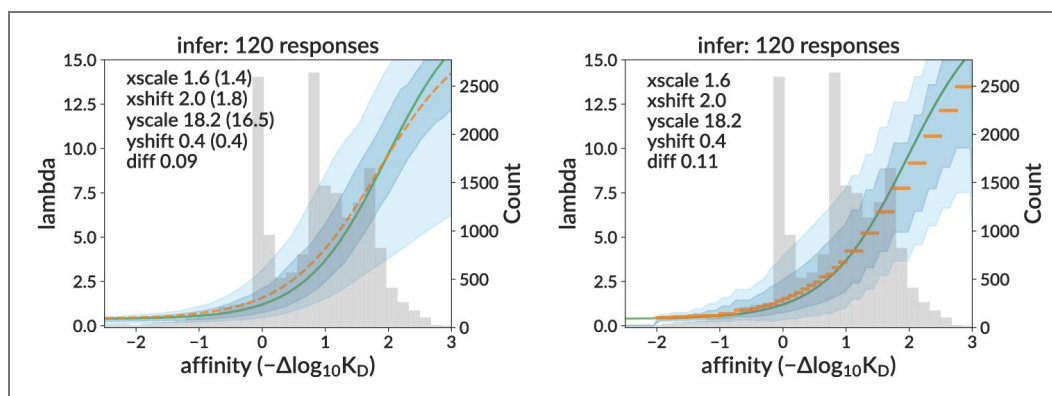


Figure 6. Inferred sigmoid curves for the central data mimic data-like simulation sample.

The medoid curve is shown in orange, and true curve in green. This sample consists of 120 GCs all simulated with the same sigmoid parameters (from the central data prediction) and non-sigmoid parameters (from the best-matched summary statistics).

Effective birth rates

The birth rate described by the response function is an absolute quantity λ that, in the absence of carrying capacity constraint, would be simply inverted to get the waiting time for birth events. With capacity constraint, however, we apply a scaling factor m (Eq 2) before inversion. It can thus be useful to scale by m and subtract the cell-specific death rate μ_i to calculate an “effective” birth rate $m\lambda_i - \mu_i$ describing the net fitness advantage of cell i at a given moment in time. We now call the original λ the “intrinsic” birth rate to distinguish it from this effective rate. We show this effective birth rate on several GCs with central data mimic simulation parameters that were allowed to persist to 50 days (far beyond the extracted GC data at 15 and 20 days, Figure 7 [↗](#)).

The most striking feature of this effective rate is its variability: rather than being a fixed property of the GC system, it describes the evolutionary micro-environment of a particular GC at a particular moment in time. Overall, the effective rate has a much smaller scale, generally staying between -0.2 and 1 , compared to 0 to more than 15 for the intrinsic rate. The effective rate is also time-dependent, with m compressing its range as time goes on, which also contrasts with the fixed intrinsic rate. This compression continues well after the simulation has reached steady state population (blue line in right column, Figure 7 [↗](#)), as the mean affinity (red line in right column) continues to move relative to the intrinsic birth rate’s upper and lower asymptotes (middle column). Biologically speaking, this compression means that there is less reward for gaining affinity later in the simulation than in the beginning. This feature is shared with the traveling wave model (DeWitt et al., 2025 [↗](#), Fig. S6D).

In order to compare more directly to DeWitt et al. (2025) [↗](#), we remade their Fig. S6D, truncating to values at which affinities are actually observed in the bulk data, and using only three of the seven timepoints (11, 20, and 70, Figure 8 [↗](#) left). We then simulated 25 GCs with central data mimic parameters out to 70 days. For each such GC, we found the time point with mean affinity over living cells closest to each of three specific “target” affinity values (0.1, 1.0, 2.0) corresponding to the mean affinity of the bulk data at timepoints 11, 20, and 70. We then plot the effective birth rates of all living cells vs relative affinity (subtracting mean affinity) at the resulting GC-specific timepoints for all 25 GCs together (Figure 8 [↗](#) right). Note that because each GC evolves at very different and time-dependent rates, we could not simply use the timepoints from the bulk data, since each GC slice from our simulation would then have very different mean affinity. The mean over GCs of these GC-specific chosen times is 10.9, 24.5, 44.4 (compared to the original bulk data time points 11, 20, 70). It is important to note that while the first two target affinities (0.1 and 1.0) are within the affinity ranges encountered in the extracted GC data, the third value (2.0) is far beyond them, and thus represents extrapolation to an affinity regime informed more by our underlying model than by the real data on which we fit it.

Despite representing fundamentally different quantities, at positive relative affinities the two estimates are relatively similar. At negative affinities, however, they diverge sharply, with the net growth rate from (DeWitt et al., 2025 [↗](#)) falling without limit while our estimate of $m\lambda - \mu$ quickly plateaus. We defer a more extensive explanation of these features to the Discussion (see reference to Figure 8 [↗](#)). In order to make the lefthand plot in Figure 8 [↗](#), we used the Python notebook from (DeWitt et al., 2025 [↗](#)) at <https://github.com/matsengrp/gcreplay/blob/main/analysis/affinity-fitness-response.ipynb> [↗](#).

Discussion

The germinal center reaction is a complex evolutionary process during which B cell populations improve their affinity. Little is known, however, about the specific form of the underlying relationship between affinity and fitness. Because a B cell’s fitness depends on the context of other cells with which it interacts, this relationship cannot be measured directly in a lab. Its form must instead be inferred based on observations of intact, evolving germinal centers.

We model this biology using a birth-death-mutation process with a carrying capacity constraint. While birth-death models are common, the necessity of a capacity constraint makes likelihood evaluation intractable using existing methods (although recent theoretical work may provide a

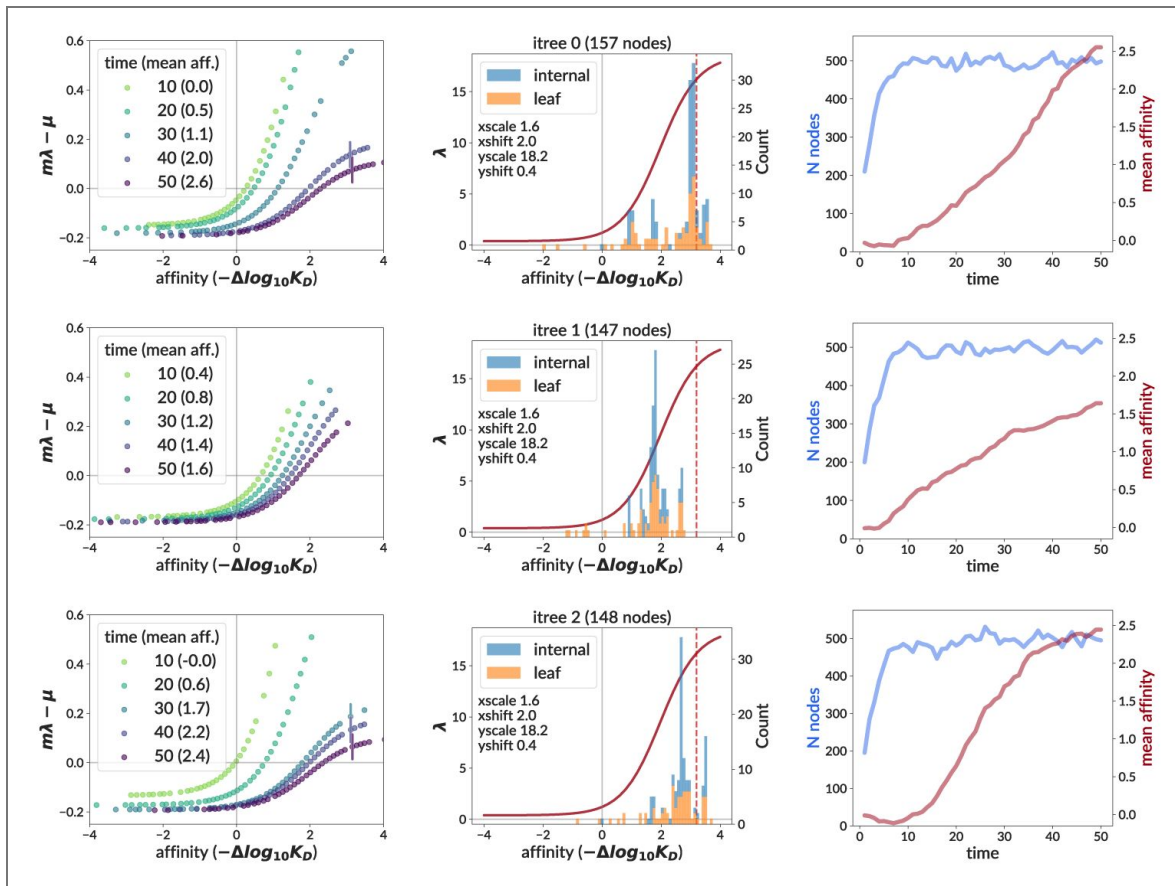


Figure 7. Effective birth rates (left column) calculated for three GCs simulated with the central data mimic parameters.

The effective birth rate for cell i is defined as $m\lambda_i - \mu_i$ for the carrying capacity modulating factor m , intrinsic birth rate λ_i , and death rate μ_i . It is shown at several different time values for all living cells, except that for plotting clarity, cells closer to each other than 0.1 in affinity are not shown. Note that we extend time here to 50 days in order to aid comparison to the bulk data used by the traveling wave model (DeWitt et al., 2025, Fig. S6D), which assumes a steady state. (Our extracted GC data is sampled at 15 and 20 days.) Vertical lines (left and middle columns) mark the point at which the slope has dropped to 1/2 its maximum value (if absent, the slope never falls below this threshold). We also show the intrinsic birth rate (middle column, solid red curve), and include histograms of sampled affinities at the final time point. The right column shows time traces of the number of living cells (i.e. nodes, in blue) and the mean affinity of all living cells (red).

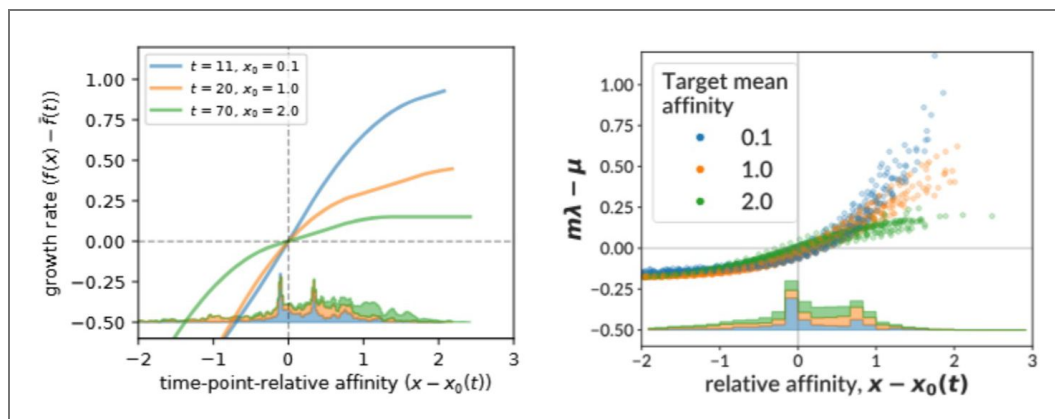


Figure 8. The net growth rate from (DeWitt et al., 2025 [DOI](#), Fig. S6D) (left) compared to our effective birth rate (right)

for three different “target” mean affinity values. We selected three well-spaced timepoints from Fig. S6D, and matched the corresponding mean affinity values from the bulk data to particular time slices in 25 GCs simulated with our central data mimic parameters (inferred on the extracted GC data) out to time 70. The net growth rate describes the net population growth at affinity x and time t in a traveling wave fitness model (see text). In our model, the effective birth rate is defined and plotted as in Figure 7 [DOI](#). We also show stacked histograms of observed affinity values from the bulk data (left) and the 25 simulated GCs (*not* from the extracted GC data) (right). The measured affinity values from the extracted GC data (which extends only to time 20) are shown in Figure 4 [DOI](#).

foundation for solving this in a “mean-field” sense as described in [DeWitt et al., 2024](#) [\(link\)](#). We thus pursue a likelihood-free approach, training a deep neural network on simulation designed to emulate the replay experiment data. This likelihood-free approach also allows us to use a model of mutation on sequences that need not be Markov on affinity “types.”

We used this neural network to infer the affinity-fitness response function. Because the model includes several parameters that are not well constrained by experiment, but whose prediction with the neural network would be infeasibly complicated, we adopted a two-step inference procedure. We first infer the sigmoid shape at many different potential non-sigmoid parameter values, then “fit” the non-sigmoid parameters using summary statistics. In testing on simulation, the network showed the ability to reliably infer the sigmoid’s shape on samples similar to the extracted GC data. When applied to that data, it inferred a consistent shape across trees that was in line with expectations from simulation, while also giving summary statistics well-matched to the data. This shape indicates that intrinsic fitness (defined as the rate of binary cell division before applying capacity constraint) roughly triples from affinity 0 (naive) to 1, and then triples again from 1 to 2. We did not observe any significant decrease in the response function slope (what we would call a ceiling on fitness) within the time range (20 days) of the GC extraction experiment, although we did see a decrease at affinities around 3.5, reached by day 50 in the data mimic simulation ([Figure 7](#) [\(link\)](#)). This latter result should be treated with caution, however: this ceiling derives from an affinity region (see [Figure 4](#) [\(link\)](#)) without observed affinities, and is thus largely reflective of our assumed sigmoid shape, rather than any direct information from data. We also note that the extracted GC data has very low levels of mutation compared to typical BCR repertoires, so a ceiling may manifest only at higher affinity values than we observe.

Prior work

Measurements relevant to the affinity-fitness relationship

A variety of prior biological measurements help guide our expectations for the shape of the response function. The function’s slope, for instance, controls the speed of evolution in the GC, with a steeper slope creating stronger clonal bursts and faster selective sweeps. We are not, however, aware of any work numerically quantifying the shape of this important curve other than our recent manuscript ([DeWitt et al., 2025](#) [\(link\)](#)). The existence and potential location of a ceiling on fitness or affinity, on the other hand, has been the subject of much work. While affinity and fitness ceilings are separate concepts, they are closely related. An affinity ceiling is a limit to affinity for a given antigen: there are no mutations that can improve affinity beyond this level. This would result in a truncated response function, undefined beyond the affinity ceiling. A fitness ceiling, on the other hand, is an upper asymptote on the response function. Such a ceiling would result in a limit on affinity for a germinal center reaction, since once cells are well into the upper asymptote of fitness they are no longer subject to selective pressure.

One study ([Batista and Neuberger, 1998](#) [\(link\)](#)) measured how the B cell response to antigen varies with affinity, finding that association constant $K_a = 1/K_D$ values above $10^{10} M^{-1}$ did not lead to increased B cell triggering. This value is roughly equal to one derived previously ([Foote and Eisen, 1995](#) [\(link\)](#)) starting from two assumptions: that diffusion places a fundamental upper limit to k_{on} (where $K_D = k_{off}/k_{on}$), and that dissociation half lives are less than one hour. Using our naive K_D of $4 \times 10^{-8} M$ ([Tas et al., 2016](#) [\(link\)](#)), the ceiling K_a value of $10^{10} M^{-1}$ corresponds to an affinity in our plots of $-\log_{10} \frac{10^{-10}}{4 \times 10^{-8}} = 2.6$, a value at which we observe few enough values in data ([Figure 4](#) [\(link\)](#)) that our inferred response function there is reflective of our assumed sigmoid shape rather than any direct information from data. Also note that the additive, DMS-based affinity measurements that we use are only validated between about -1 and 2 ([DeWitt et al., 2025](#) [\(link\)](#), [Fig. 2](#) [\(link\)](#)).

A minimum, “threshold” K_a value of $10^6 M^{-1}$, below which B cells do not respond, has also been reported ([Batista and Neuberger, 1998](#) [\(link\)](#)). While we do not attempt to model such a threshold in our simulation, this value would correspond to an affinity of $-\log_{10} \frac{10^{-6}}{4 \times 10^{-8}} = -1.4$ in our plots, which is a region in which we observe the expected flat response function, although note that we also observe very few cells here ([Figure 4](#) [\(link\)](#)).

Direct Inference of the affinity-fitness relationship

We have undertaken two other, independent efforts to infer the specific relationship between affinity and fitness, and it would be useful to compare their results. In DeWitt et al. (2025) [\[1\]](#), we applied a traveling wave fitness approach (Neher, 2013 [\[2\]](#); Nourmohammad et al., 2013 [\[3\]](#)) to the replay bulk data. Rather than, as in this paper, treating the evolution of individual cells and their BCR sequences, this models the distribution of cell fitnesses $p(x, t)$ as a traveling wave. The time evolution of $p(x, t)$ is controlled by two parameter functions: $q(x, y)$ describing the mutational flux from affinity state x to state y (derived from the DMS measurements), and the fitness landscape $f(x)$ specifying the fitness of a cell with affinity x (an arbitrary function fit as part of the inference procedure). The net rate of population growth for a cell with affinity x at time t is thus $f(x) - \overline{f(t)}$, with $\overline{f(t)}$ the mean fitness of the population at time t . This difference can be thought of as an effective birth rate analogous to our $m\lambda - \mu$ (Figure 7 [\[4\]](#)). The evolution of $p(x, t)$ is then determined by the differential equation:

$$\frac{\partial}{\partial t} p(x, t) = \underbrace{(f(x) - \overline{f(t)}) p(x, t)}_{\text{net population growth}} + \underbrace{\int_{-\infty}^{\infty} (q(y, x) p(y, t) - q(x, y) p(x, t)) dy}_{\text{net mutation to state x}} \quad (7)$$

where we see that the dynamics of the traveling wave p are given by the sum of two terms: net population growth and mutational flux.

We compared the traveling wave model's net growth rate to our effective birth rate in Figure 8 [\[5\]](#), finding similar behavior for above-average affinities, but divergent values below. While this discrepancy is certainly noteworthy, it likely reflects a difference in input assumptions rather than truly different inferences. Our sigmoid model forces a horizontal asymptote in effective birth rate whose value is fixed by the relatively plentiful values near zero relative affinity. At the more negative values where the discrepancy becomes significant, the effective birth rate simply reflects our sigmoid assumption, with little input from the relatively sparse data. While we have not performed this analysis for our per-bin model, it also depends on a sigmoid (implicitly via training data), so would likely display similar behavior. The net growth rate from the traveling wave model, on the other hand, is more flexible.

Although our birth-death model and the traveling wave model have similar intent, they have very different structures and assumptions. The traveling wave represents the population as a continuous density $p(x, t)$ over affinity space, taking a large-population, deterministic limit. It also assumes a steady state with respect to total population size, meaning the distribution $p(x, t)$ evolves but total population size does not. In contrast, we simulate individual discrete cells, each with its own sequence, affinity, and stochastic fate. We explicitly model population dynamics: growth from a small founder population toward carrying capacity, with the possibility of stochastic extinction. This distinction is particularly important early in the GC reaction, when populations are small and growing. During this phase, stochastic effects are strongest, the extinction risk is highest, and the dynamics differ qualitatively from the later quasi-steady-state regime. It is remarkable that abstracting away so many granular biological details, and simply modeling a two-dimensional probability distribution, yields a coherent picture of GC evolution; however it is far from clear that such simplification has no effect on the resulting inference. Indeed, in our model we see clear effects of stochasticity, where single mutations on single cells can quickly sweep through the population, yielding very different trajectories for different GCs. The discrepancy at below-average affinities could also be related to differing treatments of expression. Our model completely ignores it, whereas real B cells must maintain a minimal expression level in order to survive.

The clear advantage of the traveling wave model is its simplicity: if its high level view is accurate enough to effectively model the relevant GC dynamics, it is far more tractable. But reproducing low-level biological detail, and making high-dimensional real data comparisons (e.g. Figure 5 [↗](#)) to iteratively improve model fidelity, are also useful, providing direct evidence that we are correctly modeling the underlying biological processes. The two approaches also utilize different types of data: we use a single time point, and thus must reconstruct evolutionary history; whereas the traveling wave requires a series of timepoints. The availability of both types of data is a unique feature of the replay experiment, and provides us with the opportunity to directly compare the approaches.

We also note the different datasets used by the two methods. The bulk data used by the traveling wave model consists of counts of affinity values from bulk sequencing at seven time points (from 8 to 70 days), whereas our model uses phylogenetic trees with sequence and affinity annotations from single cell data extracted from GCs at two time points (15 and 20 days). Aside from the difference in available information between affinity counts and annotated trees, the longer time scale of the bulk data results in a larger range of measured affinity values. The bulk data also merges sequences from many GCs together before inference, whereas the extracted GC data analyzes each GC separately before jointly inferring results.

Both the intrinsic and effective birth rates are useful quantities, with each being appropriate for different tasks. The intrinsic rate is an unchanging, fundamental property of the GC system; but by itself it doesn't tell us how many offspring to expect for a given cell in a particular GC. The effective rate, on the other hand, tells us the expected number of offspring for each cell at each point in time, but with the tradeoff that it has no fixed form: it depends, through m (Eq 2), on the sum over cells of both birth and death rates, as well as on the number of cells alive at that time. Also note that because it depends on the particular population of cells alive at a given time, it is not defined along the entire x axis, but only at x values at which living cells reside (calculating its value at an affinity that no cells have would presuppose adding a cell at that affinity, which would modify the calculation for all other cells).

We have also worked to infer the response function using likelihoods under a birth-death model (Bakis et al., 2025 [↗](#)). This requires solutions of ordinary differential equations (ODEs) that account for unsampled lineages in birth-death models. Because these models require independent evolution of lineages, they cannot incorporate the effects of either carrying capacity or competition between cells. They also require Markov type evolution, whereas sequence evolution in the GC is not Markovian. While we thus do not expect these likelihood-based results to describe real data, it is nevertheless instructive to compare them to our effective birth rates. The main difference is the much earlier ceiling in Bakis et al. (2025) [↗](#), plateauing entirely by around $x = 0.5$. This early plateau is simply a consequence of the lack of population size constraint: without a carrying capacity, the model can only avoid unbounded population growth with an early, and low, fitness ceiling. Since capacity constraint cannot be ignored in the context of the GC, indeed providing the foundation of selection there, this likelihood-based work is best understood as a substantial mathematical advance toward a potential future method that could infer response functions on real data.

Recent work has developed birth-death processes with mean-field interactions (DeWitt et al., 2024 [↗](#)) that could be used for inference under models with population size constraint. Although the theoretical promise for such models is clear, inference has not yet been implemented. In future work it would be interesting to compare these methods with those presented in this manuscript.

Limitations

Our work is subject to a number of limitations, most importantly related to our model and simulation. As noted above, several aspects of our model's organization may not accurately reflect real GC dynamics. Furthermore, even for aspects that are accurately modeled, chosen parameter values may either not reflect true biological values, or may represent subtly different "effective"

parameters and thus also need modification. Our simulation also does not include recent results showing that the mutation rate may be depressed after a strong light-zone signal (Pae et al., 2025a [↗](#)), or that lower affinity cells may be subject to higher SHM rates (Li et al., 2025 [↗](#)).

Other possible misspecifications involve dark-zone/light-zone cycling and cell sampling. Our simulation implements a single-generation proliferative phase with mutation (dark zone) always followed by a single, separate selection phase (light zone). However, this may not accurately represent real GC dynamics; for instance this cannot simulate multiple proliferative phases between each selection phase (“clonal bursts”) (Tas et al., 2016 [↗](#); Pae et al., 2025b [↗](#)). It is also known that some selection occurs in the dark zone (Mayer et al., 2017 [↗](#)), at least against nonfunctional cells, and our two-category death rate may not be sufficient to model this accurately. We also sample uniform randomly from all living cells at the final time point, but it is possible that in real GCs cell sampling is based on more biased processes, for instance if higher affinity cells, or cells in a particular zone, are more likely to be sampled.

We also neglect competition among lineages stemming from different rearrangement events (different clonal families), instead assuming that each GC is seeded with instances of only a single naive sequence, and that neither cells nor antibodies migrate between different GCs. More realistically for the polyclonal GC case, we would allow lineages stemming from different naive sequences to compete with each other both within and between GCs (Zhang et al., 2013 [↗](#); McNamara et al., 2020 [↗](#); Barbulescu et al., 2025 [↗](#)). Implementing competition among several clonal families within a single GC would be conceptually simple and computationally practical in our current software framework. Competition among many GCs, however, would be computationally prohibitive because our time required is primarily determined by the total population size, since at each step we must iterate over every node and every event type in order to find the shortest waiting time. For the monoclonal replay experiment specifically, however, all naive sequences are the same and so the current modeling framework is sufficient.

We also assume a single, constant death rate for functional cells. While we infer this rate with our summary statistic matching step, if the real rate depends on cell properties (other than simply being functional), the results will be incorrect. Because birth and death rates are interrelated mechanistically through phenomena such as tonic signaling (Long et al., 2022 [↗](#)) and BCR expression-dependent birth rates (Lam et al., 1997 [↗](#)), it is also unclear how much sensitivity we have to infer them independently. We note in passing that identifiability of birth and death rates from phylogenies is inherently difficult (Morlon et al., 2022 [↗](#)).

It is also possible that our choice of a sigmoid for the response function is not sufficiently flexible. This form cannot, for instance, accommodate either a decrease in fitness with affinity, or a discontinuous change in slope. It also requires symmetry around its midpoint. Because both the sigmoid and per-bin models were trained on sigmoid simulation, they can only infer sigmoid-like response functions.

While we chose to use a birth-modulated carrying capacity, it is possible that this does not accurately describe GC dynamics. We have, for instance, experimented with two other methods of capacity enforcement, one that modulates death, rather than birth, using the inverse of 2. The other, “hard”, constraint immediately kills a random individual whenever a birth event causes the population to exceed the carrying capacity.

Our summary statistic distributions are not always well-matched to data. The worst discrepancies are inaccurate structure (extra peaks) in the simulation affinity distributions, and deficits in the tails of the simulation abundance plots. The extra affinity peaks could be from inaccuracies in the sequence-affinity mapping, which assumes zero epistasis. Meanwhile, enlarging the abundance tail is easily accomplished by increasing burstiness with a steeper response function, or by decreasing diversity via either smaller initial population or smaller carrying capacity. These changes would also, of course, affect other distributions, with for instance a steeper response function increasing the speed of evolution and shifting the affinity distribution rightwards (while a smaller population does the opposite). While we can thus move the worst discrepancies from

one distribution to another, our inability to match all distributions simultaneously is probably due either to one of the model misspecifications described in the preceding paragraphs, or to a neural network inference problem leading to incorrect response function.

Our neural network could also be improved. While the CBLV encoding incorporates a number of improvements over previous schemes, the process of encoding trees for digestion by neural networks is still a very new problem, and current methods are likely not particularly close to optimal. Recent work, for instance, has improved on the CBLV scheme by incorporating the generational context of each node in a way that makes it easier for the network to understand (Perez and Gascuel, 2024 [DOI](#)). This work also introduces a network architecture that is specifically designed around the new encoding. In the future, we plan to explore the use of this new method.

The replay experiment also represents a very special case among typical BCR repertoires. While the basic dynamics at play are likely to be representative of those in wild repertoires, it is important to keep in mind that these results only reflect mutation resulting from the exposure of a single naive antibody to a single experimental antigen. Some aspects of behavior in the low-SHM/early times regime of the extracted GC data are also potentially different to those at the higher SHM levels and longer times found in typical repertoires. This is especially relevant to affinity or fitness ceilings, to which we likely have little sensitivity with the current data. Additionally, repeating the replay experiment using a less-optimized antigen with lower starting affinity might enable us to better map out dynamics during the early stages of affinity maturation when, for example, specificity may have some plasticity (Zuo et al., 2025 [DOI](#)).

Conclusion and outlook

In closing, simulation-based methods have enabled inference on this otherwise-intractable, messy biological problem. However, they were not a panacea. Simulation-based inference is difficult when the model's behavior is determined by many parameters, since the product of many dimensions cannot be tested exhaustively. The optimality surface was also found to be quite rugged: it was easy to find combinations of parameters that lead to pathologies. Previous successful studies using simulation-based inference on trees have had fewer parameters and a less complex objective (Voznica et al., 2022 [DOI](#); Thompson et al., 2024 [DOI](#)). Nevertheless, we were able to infer parameters of a complex and dynamic process that is hidden inside the germinal center.

Looking forward, even if by some miracle we could perform inference under an arbitrary forwardtime model, interesting biological questions would remain. For instance, what mechanism limits the population size of the germinal center? In this paper we chose birth-limited population size control, and the validity of our results likely depends somewhat on this assumption. In comparison with the traveling wave model of DeWitt et al. (2025) [DOI](#), which has a separate structure and different assumptions, we could see that the inferred properties differed. Finding a convincing answer to this question will require additional experiments and likely additional modeling.

Figure supplements

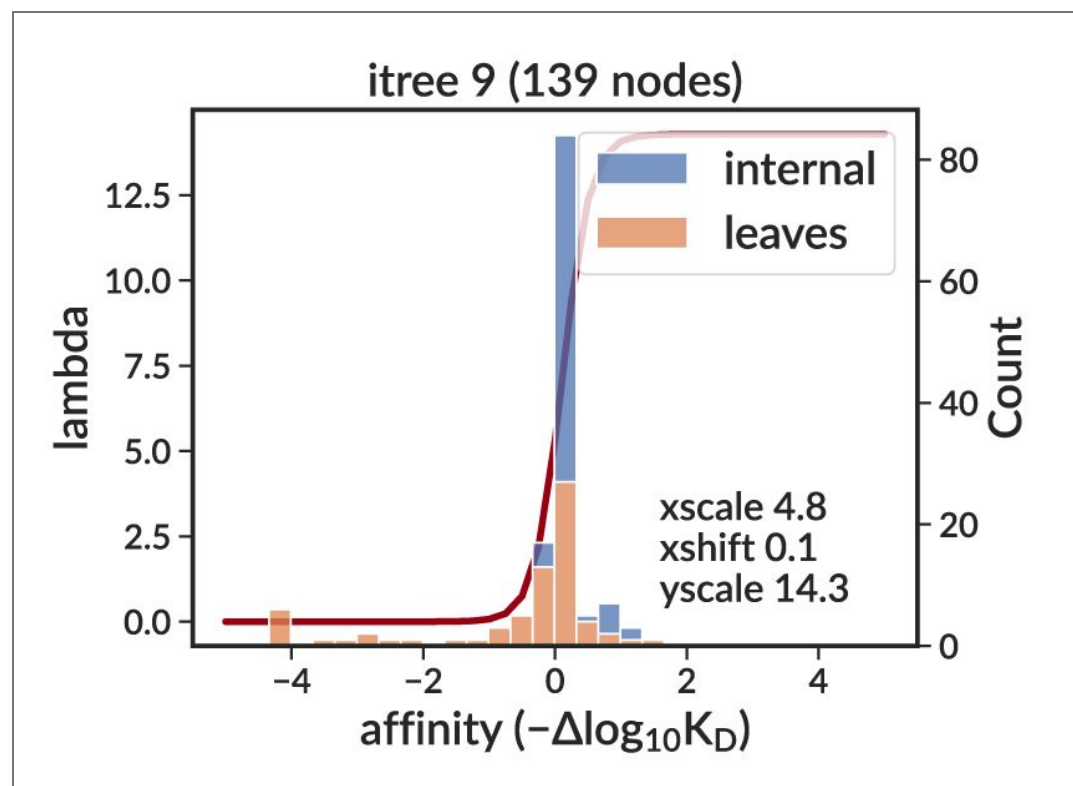


Figure 1—figure supplement 2. Curve difference loss function calculation.

We divide the “difference” area (red bars) between the true (green) and inferred (red) curves by the area (light red) under the true curve within the bounds $[-2.5, 3]$. The loss value is thus the fraction of the area under the true curve represented by the area between the true and inferred curves.

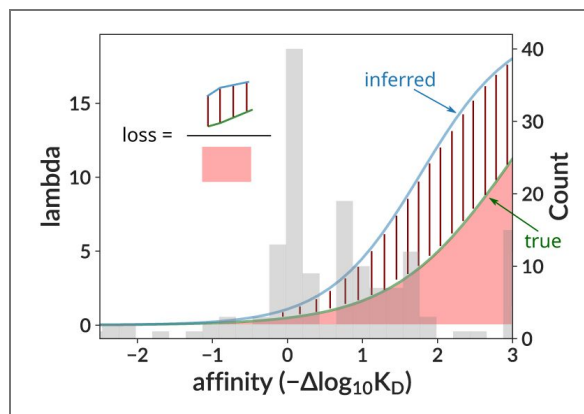
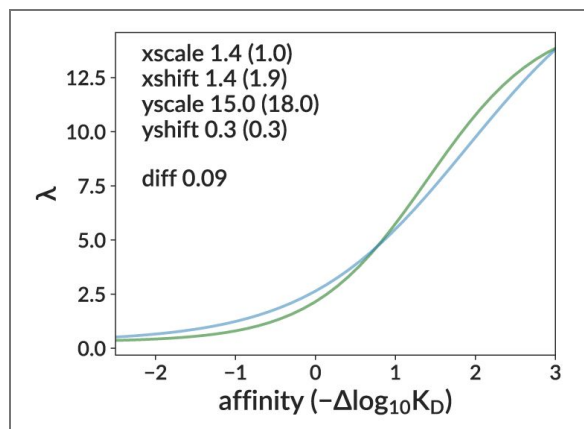


Figure 1—figure supplement 3. Example of approximate sigmoid parameter degeneracy.

A potential pair of true (green) and inferred (blue) sigmoid curves, with true (inferred) parameter values. In the range shown, the inferred curve has compensated for a too-small xscale (transition steepness) by increasing xshift (shifting to the right) and yscale (upper asymptote). This results in a curve difference loss value of only 9%, which is much smaller our expected inference accuracy.



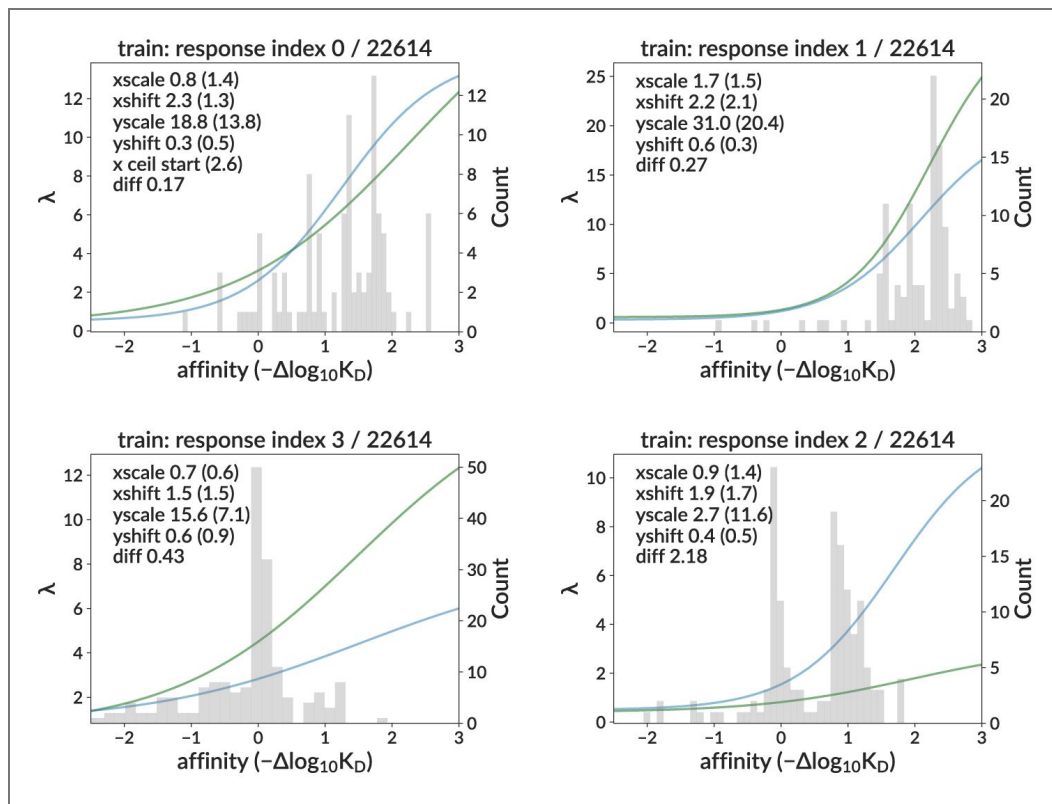


Figure 2—figure supplement 1. Example true and inferred response functions on the training sample.

The “diff” entry in the per-plot table gives the curve difference loss.

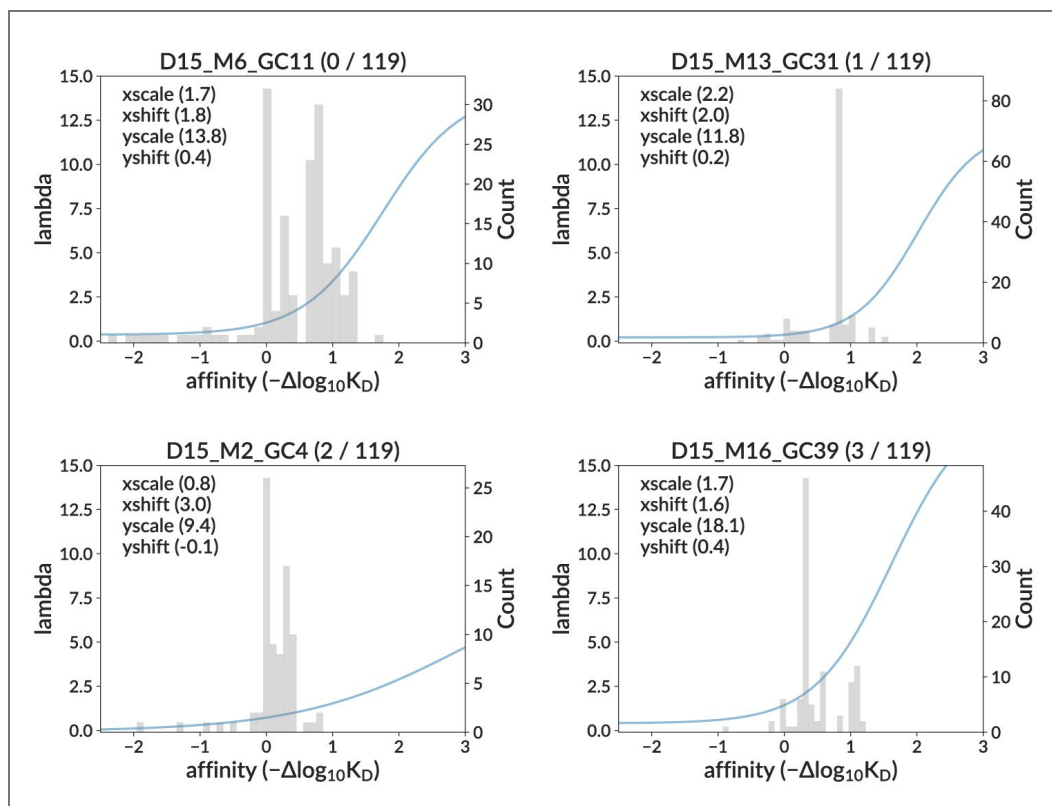


Figure 4—figure supplement 1. Example inferred sigmoid curves on data

for four representative GCs.

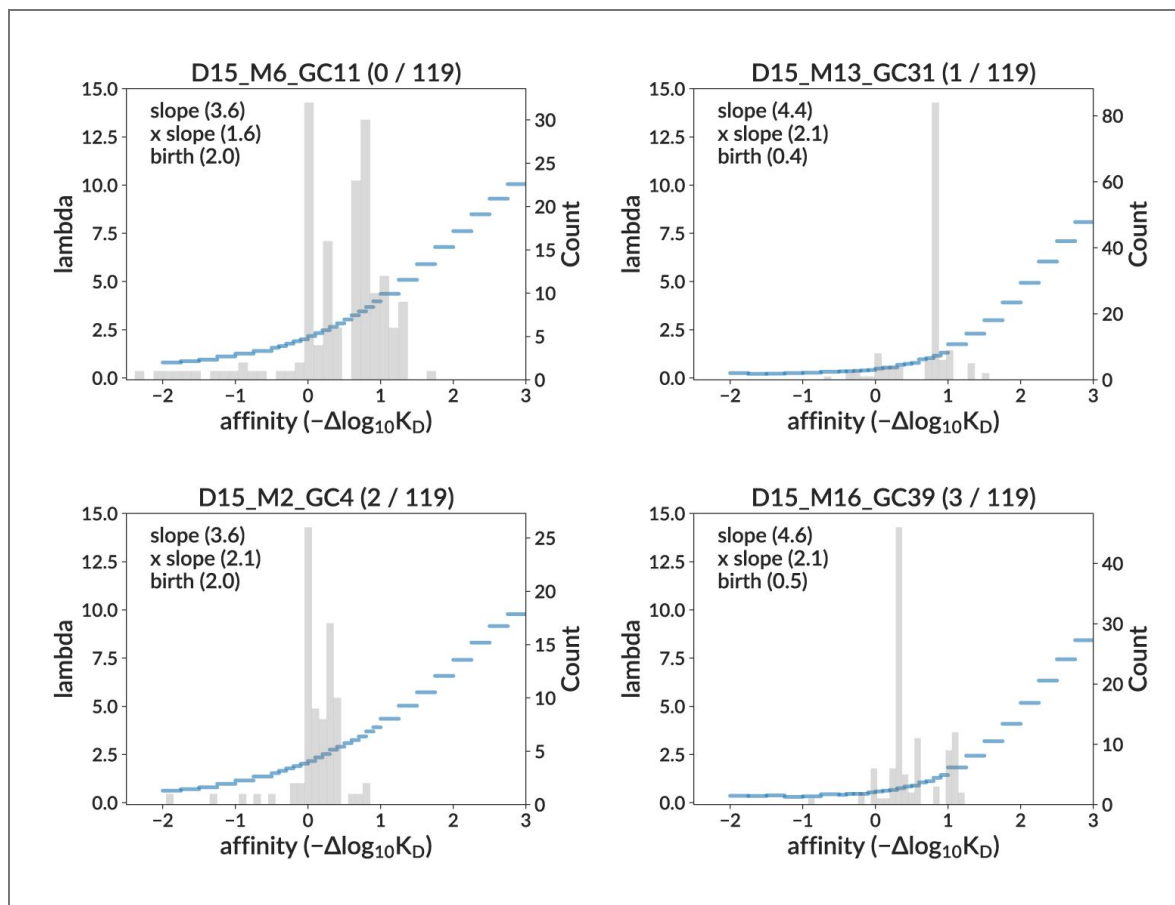


Figure 4—figure supplement 2. Example inferred per-bin response functions on data for four representative GCs.

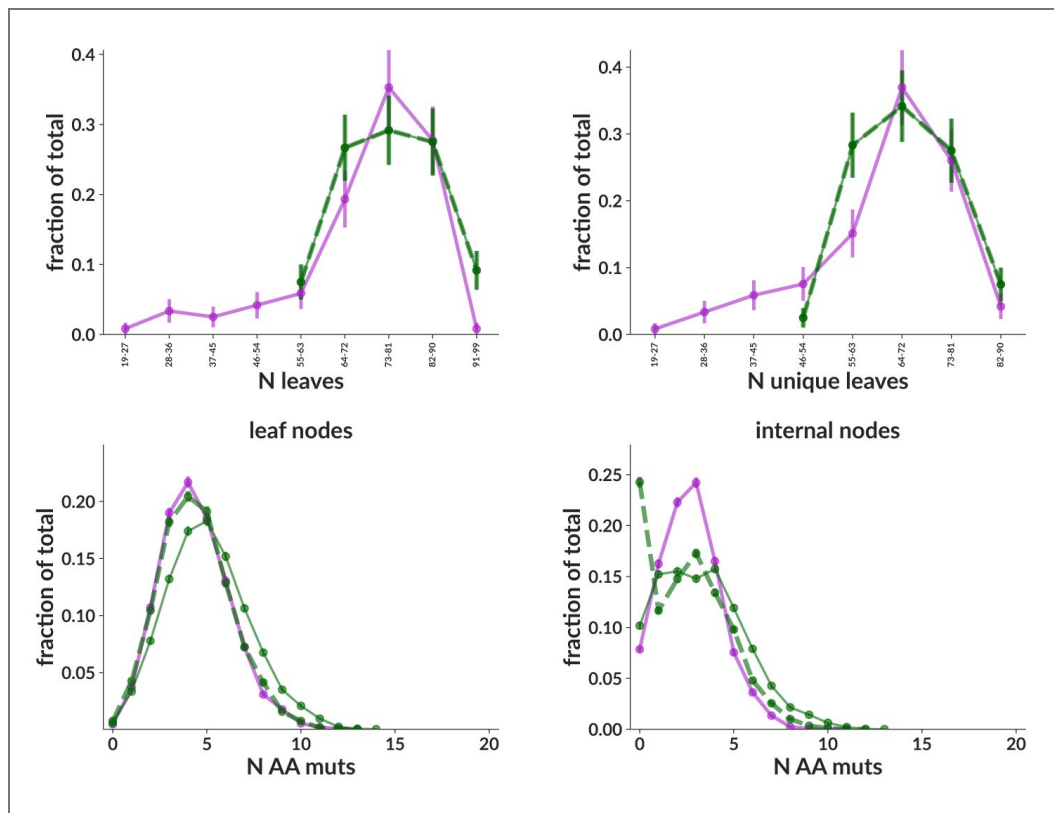


Figure 5—figure supplement 1. Additional summary statistics distributions for the same central data mimic sample as the main figure.

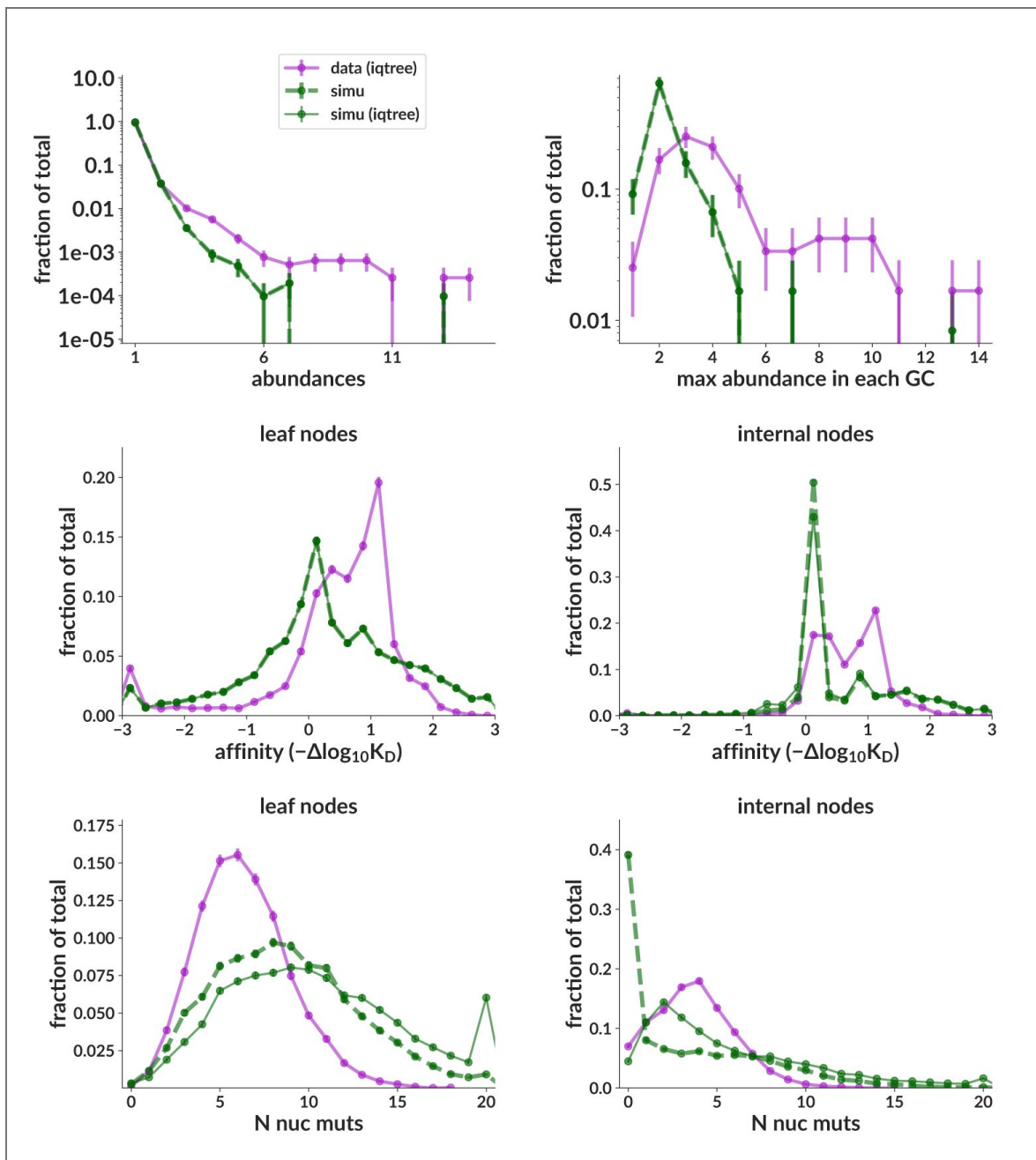


Figure 5—figure supplement 2. Summary statistic distributions for the simulation sample used for training.

In order to allow easy comparison of abundance distributions, these plots include only 120 of the simulated trees. This sample is designed to have a wide range of parameters, encompassing all plausible true data values, and is thus not designed to closely match data summary statistics.

Data availability

All raw and processed real and simulated data samples are available at:

<https://doi.org/10.5281/zenodo.15022130>  All code is available at:

<https://github.com/matsengrp/gcdyn> 

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Additional information

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References

- Adams RM**, Mora T, Walczak AM, Kinney JB (2016) Measuring the sequence-affinity landscape of antibodies with massively parallel titration curves. *eLife* **5** <https://doi.org/10.7554/eLife.23156> | <https://doi.org/10.7554/eLife.23156>
- Allen D**, Simon T, Sablitzky F, Rajewsky K, Cumano A (1988) Antibody engineering for the analysis of affinity maturation of an anti-hapten response. *EMBO J* **7**:1995-2001
- Bakis AG**, Vora AA, Araki T, Jia T, Galloway JG, Jennings-Shaffer C, Victora GD, Song YS, DeWitt WS, Matsen FA, *et al.* (2025) Bayesian inference of antibody evolutionary dynamics using multitype branching processes. *arXiv*
- Barbulescu A**, Bilanovic J, Langelaar T, Teetz AK, Urnavicius L, Hobbs A, Shen JJ, Abrahamse NH, de Carvalho RVH, Mesin L, *et al.* (2025) Antibody-mediated feedback modulates interclonal competition in the germinal center. *bioRxiv*
- Barido-Sottani J**, Vaughan TG, Stadler T (2020) A Multitype Birth-Death Model for Bayesian Inference of Lineage-Specific Birth and Death Rates. *Syst Biol* **69**:973-986 <https://doi.org/10.1093/sysbio/syaa016> | <https://doi.org/10.1093/sysbio/syaa016>
- Batista FD**, Neuberger MS (1998) Affinity dependence of the B cell response to antigen: a threshold, a ceiling, and the importance of off-rate. *Immunity* **8**:751-759 [https://doi.org/10.1016/s1074-7613\(00\)80580-4](https://doi.org/10.1016/s1074-7613(00)80580-4)
- Beaumont MA**, Zhang W, Balding DJ (2002) Approximate Bayesian Computation in Population Genetics. *Genetics* **162**:2025-2035 <http://www.genetics.org/content/162/4/2025>
- Berek C**, Milstein C (1987) Mutation drift and repertoire shift in the maturation of the immune response. *Immunol Rev* **96**:23-41

- Budd GE, Mann RP** (2018) History is written by the victors: The effect of the push of the past on the fossil record: PERSPECTIVE. *Evolution* **72**:2276-2291 <https://doi.org/10.1111/evo.13593>
- Burton DR, Ahmed R, Barouch DH, Butera ST, Crotty S, Godzik A, Kaufmann DE, McElrath MJ, Nussenzweig MC, Pulendran B, et al.** (2012) A blueprint for HIV vaccine discovery. *Cell Host Microbe* **12**:396-407
- Cui A, Di Niro R, Vander Heiden JA, Briggs AW, Adams K, Gilbert T, O'Connor KC, Vigneault F, Shlomchik MJ, Kleinstein SH** (2016) A Model of Somatic Hypermutation Targeting in Mice Based on High-Throughput Ig Sequencing Data. *J Immunol* **197**:3566-3574 <https://doi.org/10.4049/jimmunol.1502263> | <https://doi.org/10.4049/jim-munol.1502263>
- Cyster JG, Allen CDC** (2019) B Cell Responses: Cell Interaction Dynamics and Decisions. *Cell* **177**:524-540
- Dessalles R, D'Orsogna M, Chou T** (2018) Exact steady-state distributions of multispecies birth-death-immigration processes: Effects of mutations and carrying capacity on diversity. *J Stat Phys* **173**:182-221 <https://doi.org/10.1007/s10955-018-2128-4> | <https://doi.org/10.1007/s10955-018-2128-4>
- DeWitt WS, Evans SN, Hiesmayr E, Hummel S** (2024) Mean-field interacting multi-type birth-death processes with a view to applications in phylodynamics. *Theor Popul Biol* **159**:1-12 <https://doi.org/10.1016/j.tpb.2024.07.002> | <https://doi.org/10.1016/j.tpb.2024.07.002>
- DeWitt WS, Vora AA, Araki T, Galloway JG, Alkutar T, Bortolatto J, Castro TBR, Dumm W, Jennings-Shaffer C, Jia T, et al.** (2025) Replaying germinal center evolution on a quantified affinity landscape. *bioRxiv*
- Foote J, Eisen HN** (1995) Kinetic and affinity limits on antibodies produced during immune responses. *Proc Natl Acad Sci USA* **92**:1254-1256
- Jacobsen JT, Mesin L, Markoulaki S, Schiepers A, Cavazzoni CB, Bousbaine D, Jaenisch R, Victora GD** (2018) One-step generation of monoclonal B cell receptor mice capable of isotype switching and somatic hypermutation. *J Exp Med* **215**:2686-2695
- Kuraoka M, Schmidt AG, Nojima T, Feng F, Watanabe A, Kitamura D, Harrison SC, Kepler TB, Kelsoe G** (2016) Complex Antigens Drive Permissive Clonal Selection in Germinal Centers. *Immunity* <https://doi.org/10.1016/j.immuni.2016.02.010> | <https://doi.org/10.1016/j.immuni.2016.02.010>
- Kühnert D, Stadler T, Vaughan TG, Drummond AJ** (2016) Phylodynamics with Migration: A Computational Framework to Quantify Population Structure from Genomic Data. *Mol Biol Evol* **33**:2102-2116 <https://doi.org/10.1093/molbev/msw064> | <https://doi.org/10.1093/molbev/msw064>
- Lajaaiti I, Lambert S, Voznica J, Morlon H, Hartig F** (2023) A Comparison of Deep Learning Architectures for Inferring Parameters of Diversification Models from Extant Phylogenies. *bioRxiv* <https://doi.org/10.1101/2023.03.03.530992>
- Lam KP, Kühn R, Rajewsky K** (1997) In vivo ablation of surface immunoglobulin on mature B cells by inducible gene targeting results in rapid cell death. *Cell* **90**:1073-1083
- Lambert S, Voznica J, Morlon H** (2023) Deep Learning from Phylogenies for Diversification Analyses. *Syst Biol* <https://doi.org/10.1093/sysbio/syad044> | <https://doi.org/10.1093/sysbio/syad044>
- Landis MJ, Thompson A** (2025) phyddle: software for exploring phylogenetic models with deep learning. *Systematic Biology* syaf036
- Li R, Bao K, Liu C, Ma X, Hua Z, Zhu P, Hou B** (2025) Competition propels, rather than limits, the success of low-affinity B cells in the germinal center response. *Cell Rep* **44**:115334
- Long Z, Phillips B, Radtke D, Meyer-Hermann M, Bannard O** (2022) Competition for refueling rather than cyclic reentry initiation evident in germinal centers. *Sci Immunol* **7**:eabm0775
- Mayer CT, Gazumyan A, Kara EE, Gitlin AD, Golijanin J, Viant C, Pai J, Oliveira TY, Wang Q, Escolano A, et al.** (2017) The microanatomic segregation of selection by apoptosis in the germinal center. *Science* **358**:eaao2602

- McNamara HA**, Idris AH, Sutton HJ, Vistein R, Flynn BJ, Cai Y, Wiehe K, Lyke KE, Chatterjee D, Kc N, *et al.* (2020) Antibody feedback limits the expansion of B cell responses to malaria vaccination but drives diversification of the humoral response. *Cell Host Microbe* **28**:572–585.e7
- Mesin L**, Schiepers A, Ersching J, Barbulescu A, Cavazzoni CB, Angelini A, Okada T, Kurosaki T, Victora GD (2020) Restricted Clonality and Limited Germinal Center Reentry Characterize Memory B Cell Reactivation by Boosting. *Cell* **180**:92–106.e11
- Minh BQ**, Schmidt HA, Chernomor O, Schrempf D, Woodhams MD, von Haeseler A, Lanfear R (2020) IQ-TREE 2: New models and efficient methods for phylogenetic inference in the genomic era. *Mol Biol Evol* **37**:1530–1534
- Morlon H**, Robin S, Hartig F (2022) Studying speciation and extinction dynamics from phylogenies: addressing identifiability issues. *Trends Ecol Evol* **37**:497–506
<https://doi.org/10.1016/j.tree.2022.02.004> | <https://doi.org/10.1016/j.tree.2022.02.004>
- Murugan R**, Buchauer L, Triller G, Kreschel C, Costa G, Pidelaserra Martí G, Imkeller K, Busse CE, Chakravarty S, Sim BKL, *et al.* (2018) Clonal selection drives protective memory B cell responses in controlled human malaria infection. *Sci Immunol* **3** <https://doi.org/10.1126/sciimmunol.aap8029> | <https://doi.org/10.1126/sciimmunol.aap8029>
- Nee S**, Holmes EC, May RM, Harvey PH (1994) Extinction rates can be estimated from molecular phylogenies. *Philos Trans R Soc Lond B Biol Sci* **344**:77–82 <https://doi.org/10.1098/rstb.1994.0054> | <https://doi.org/10.1098/rstb.1994.0054>
- Neher RA** (2013) Genetic Draft, Selective Interference, and Population Genetics of Rapid Adaptation. *Annu Rev Ecol Syst* **44**:195–215 <https://doi.org/10.1146/annurev-ecolsys-110512-135920> | <https://doi.org/10.1146/annurev-ecolsys-110512-135920>
- Nourmohammad A**, Schiffels S, Lässig M (2013) Evolution of molecular phenotypes under stabilizing selection. *J Stat Mech* **2013**:P01012 <https://doi.org/10.1088/1742-5468/2013/01/P01012>
- Pae J**, Schwan N, Ottino-Löffler B, DeWitt WS, Garg A, Bortolatto J, Vora AA, Shen JJ, Hobbs A, Castro TBR, *et al.* (2025) Transient silencing of hypermutation preserves B cell affinity during clonal bursting. *Nature* 1–9
- Pae J**, Schwan N, Ottino-Löffler B, DeWitt WS, Garg A, Bortolatto J, Vora AA, Shen JJ, Hobbs A, Castro TBR, *et al.* (2025) Transient silencing of hypermutation preserves B cell affinity during clonal bursting. *Nature* 1–9 <https://doi.org/10.1038/s41586-025-08687-8>
- Perez MF**, Gascuel O (2024) PhyloCNN: Improving tree representation and neural network architecture for deep learning from trees in phylodynamics and diversification studies. *bioRxiv*
- Schrider DR**, Kern AD (2018) Supervised Machine Learning for Population Genetics: A New Paradigm. *Trends Genet* **34**:301–312 <https://doi.org/10.1016/j.tig.2017.12.005> | <https://doi.org/10.1016/j.tig.2017.12.005>
- Schwickert TA**, Lindquist RL, Shakhar G, Livshits G, Skokos D, Kosco-Vilbois MH, Dustin ML, Nussenzweig MC (2007) In vivo imaging of germinal centres reveals a dynamic open structure. *Nature* **446**:83–87
- Sheehan S**, Song YS (2016) Deep Learning for Population Genetic Inference. *PLoS Comput Biol* **12**:e1004845 <https://doi.org/10.1371/journal.pcbi.1004845>
- Suchard MA**, Lemey P, Baele G, Ayres DL, Drummond AJ, Rambaut A (2018) Bayesian phylogenetic and phylodynamic data integration using BEAST 1.10. *Virus Evol* **4**:vey016
- Tas JMJ**, Mesin L, Pasqual G, Targ S, Jacobsen JT, Mano YM, Chen CS, Weill JC, Reynaud CA, Browne EP, *et al.* (2016) Visualizing antibody affinity maturation in germinal centers. *Science* **351**:1048–1054
- Thompson A**, Liebeskind B, Scully EJ, Landis M (2024) Deep learning and likelihood approaches for viral phylogeography converge on the same answers whether the inference model is right or wrong. *Syst Biol* <https://doi.org/10.1093/sysbio/syad074> | <https://doi.org/10.1093/sysbio/syad074>
- Victora GD**, Nussenzweig MC (2022) Germinal centers. *Annu Rev Immunol* **40**:413–442

Voznica J, Zhukova A, Boskova V, Saulnier E, Lemoine F, Moslonka-Lefebvre M, Gascuel O (2022) Deep learning from phylogenies to uncover the epidemiological dynamics of outbreaks. *Nat Commun* **13**:3896 <https://doi.org/10.1038/s41467-022-31511-0> | <https://doi.org/10.1038/s41467-022-31511-0>

Zhang Y, Meyer-Hermann M, George LA, Figge MT, Khan M, Goodall M, Young SP, Reynolds A, Falciani F, Waisman A, *et al.* (2013) Germinal center B cells govern their own fate via antibody feedback. *J Exp Med* **210**:457-464

Zuo T, Gautam A, Saghaei S, Khobragade SN, Ahmed R, Mahdavinia A, Zarghami M, Pacheco GA, Green K, Travers M, *et al.* (2025) Somatic hypermutation unlocks antibody specificities beyond the primary repertoire. *Immunity*

Peer reviews

Reviewer #1 (Public review):

Summary:

This paper aims to characterize the relationship between affinity and fitness in the process of affinity maturation. To this end, the authors develop a model of germinal center reaction and a tailored statistical approach, building on recent advances in simulation-based inference.

The model provides a framework for linking affinity measurements and sequence evolution and does so while accounting for the stochasticity inherent to the germinal center reaction. The model's sophistication comes at the cost of numerous parameters and leads to intractable likelihood, which are the primary challenges addressed by the authors. The approach to inference is innovative and relies on training a neural network on extensive simulations of trajectories from the model.

The revised methods section is easier to follow and better explains the approach. Inference results on simulated data are compelling and the real-data findings are compared with alternative approaches, clarifying the relationship to previous work.

<https://doi.org/10.7554/eLife.108880.2.sa2>

Reviewer #2 (Public review):

Summary:

This paper presents a new approach for explicitly transforming B cell receptor affinity into evolutionary fitness in the germinal center. It demonstrates the feasibility of using likelihood-free inference to study this problem and demonstrates how effective birth rates appear to vary with affinity in real-world data.

Strengths:

- The authors leverage the unique data they have generated for a separate project to provide novel insights to a fundamental question.
- The paper is clearly written, with accessible methods and straightforward discussion of the limits of this model.
- Code and data are publicly available and well-documented.

Weaknesses:

- No substantial weaknesses noted.

<https://doi.org/10.7554/eLife.108880.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

Summary:

This paper aims to characterize the relationship between affinity and fitness in the process of affinity maturation. To this end, the authors develop a model of germinal center reaction and a tailored statistical approach, building on recent advances in simulation-based inference. The potential impact of this work is hindered by the poor organization of the manuscript. In crucial sections, the writing style and notations are unclear and difficult to follow.

We thank the reviewer for their kind words, and have endeavored to address all of their concerns as to the structure and style of the manuscript.

Strengths:

The model provides a framework for linking affinity measurements and sequence evolution and does so while accounting for the stochasticity inherent to the germinal center reaction. The model's sophistication comes at the cost of numerous parameters and leads to intractable likelihood, which are the primary challenges addressed by the authors. The approach to inference is innovative and relies on training a neural network on extensive simulations of trajectories from the model.

Weaknesses:

The text is challenging to follow. The descriptions of the model and the inference procedure are fragmented and repetitive. In the introduction and the methods section, the same information is often provided multiple times, at different levels of detail.

Thank you for pointing this out. We have rearranged the methods in order to make the presentation more linear, and to reduce duplication with the introduction.

Specifically, we moved the affinity definition to the start, removed the redundant bullet point list, and moved the parameter value table to the end.

This organization sometimes requires the reader to move back and forth between subsections (there are multiple non-specific references to "above" and "below" in the text).

This is a great point, we have either removed or replaced all references to "above" or "below" with more specific citations.

The choice of some parameter values in simulations appears arbitrary and would benefit from more extensive justification. It remains unclear how the "significant uncertainty" associated with these parameters affects the results of inference.

We have clarified where various parameter values come from:

"In addition to the four sigmoid parameters, which we infer directly, there are other parameters in Table 1 about which we have incomplete information. The carrying capacity method and the choice of sigmoid for the response function represent fundamental model assumptions. We also fix the death rate for nonfunctional (stop) sequences, which would be

very difficult to infer with the present experiment. For others, we know precise values from the replay experiment for each GC (time to sampling, # sampled cells/GC), but use a somewhat wider range for the sake of generalizability. The mutability multiplier is a heuristic factor used to match the SHM distributions to data. The naive birth rate is determined by the sigmoid parameters, but has its own range in order to facilitate efficient simulation.

For two of the three remaining parameters (carrying capacity and initial population), we can ostensibly choose values based on the replay experiment. These values carry significant uncertainty, however, partly due to inherent experimental uncertainty, but also because they may represent different biological quantities to those in simulation. For instance, an experimental measurement of the number of B cells in a germinal center might appear to correspond closely to simulation carrying capacity. However if germinal centers are not well mixed, such that competition occurs only among nearby cells, the "effective" carrying capacity that each cell experiences could be much smaller.

Fortunately, in addition to the neural network inference of sigmoid parameters, we have another source of information that we can use to infer non-sigmoid parameters: summary statistic distributions. We can use the matching of these distributions to effectively fit values for these additional unknown parameters. We also include the final parameter, the functional death rate, in these non-sigmoid inferred parameters, although it is unconstrained by the replay experiment, and it is unclear whether it is uniquely identifiable."

In addition, the performance of the inference scheme on simulated data is difficult to evaluate, as the reported distributions of loss function values are not very informative.

We thought of two different interpretations for this comment, so have worked to address both.

First, the comment could have been that the distribution of loss functions on the training sample does not appear to be informative of performance on data-like samples. This is true, and in our revision we have emphasized the distinction between the two types of simulation sample: those for training, where each simulated GC has different (sampled) parameter values; vs the "data mimic" samples where all GCs have identical parameters. Since the former have different values for each GC, we can only plot many inferred curves together on the latter. We also would like to emphasize that the inference problem for one GC will have much more uncertainty than will that for an ensemble of GCs (as in the full replay experiment).

"After building and training our neural network, we evaluate its performance on subsets of the training sample. While this evaluation provides an important baseline and sanity check, it is important to note that the training sample differs dramatically from real data (and the "data mimic" simulation sample that mimics real data). While real data consists of 119 GCs with identical parameters and thus response functions, we need the GCs in our training sample to span the space of all plausible parameter values. This means that while we must evaluate performance on individual GCs in the training and testing samples, in real data (and data mimic simulation) we combine results from 119 curves into a central (medoid) curve. Inference on the training sample will thus appear vastly noisier than on real data and data mimic simulation, and also cannot be plotted with all true and inferred curves together."

A second interpretation was that the reviewer did not have an intuitive sense of what a loss function value of, say, 1.0 actually means. To address this second interpretation, we have also added a supplement to Figure 2 with several example true and inferred response functions from the training sample, with representative loss values spanning 0.17 to 2.18. We have also added the following clarification to the caption of Figure 1-figure supplement 2:

"The loss value is thus the fraction of the area under the true curve represented by the area between the true and inferred curves."

Finally, the discussion of the similarities and differences with an alternative approach to this inference problem, presented in Dewitt et al. (2025), is incomplete.

We have expanded this section of the manuscript, and added a new plot directly comparing the methods.

“In order to compare more directly to DeWitt et al. 2025, we remade their Fig.S6D, truncating to values at which affinities are actually observed in the bulk data, and using only three of the seven timepoints (11, 20, and 70, Figure 8, left). We then simulated 25 GCs with central data mimic parameters out to 70 days. For each such GC, we found the time point with mean affinity over living cells closest to each of three specific “target” affinity values (0.1, 1.0, 2.0) corresponding to the mean affinity of the bulk data at timepoints 11, 20, and 70. We then plot the effective birth rates of all living cells vs relative affinity (subtracting mean affinity) at the resulting GC-specific timepoints for all 25 GCs together Figure 8, right). Note that because each GC evolves at very different and time-dependent rates, we could not simply use the timepoints from the bulk data, since each GC slice from our simulation would then have very different mean affinity. The mean over GCs of these GC-specific chosen times is 10.9, 24.5, 44.4 (compared to the original bulk data time points 11, 20, 70). It is important to note that while the first two target affinities (0.1 and 1.0) are within the affinity ranges encountered in the extracted GC data, the third value (2.0) is far beyond them, and thus represents extrapolation to an affinity regime informed more by our underlying model than by the real data on which we fit it.”

Reviewer #2 (Public review):

Summary:

This paper presents a new approach for explicitly transforming B-cell receptor affinity into evolutionary fitness in the germinal center. It demonstrates the feasibility of using likelihood-free inference to study this problem and demonstrates how effective birth rates appear to vary with affinity in real-world data.

Strengths:

(1) The authors leverage the unique data they have generated for a separate project to provide novel insights into a fundamental question. (2) The paper is clearly written, with accessible methods and a straightforward discussion of the limits of this model. (3) Code and data are publicly available and well documented.

Weaknesses (minor):

(1) Lines 444-446: I think that "affinity ceiling" and "fitness ceiling" should be considered independent concepts. The former, as the authors ably explain, is a physical limitation. This wouldn't necessarily correspond to a fitness ceiling, though, as Figure 7 shows. Conversely, the model developed here would allow for a fitness ceiling even if the physical limit doesn't exist.

Right, whoops, good point. We've rearranged the discussion to separate the concepts, for instance:

“While affinity and fitness ceilings are separate concepts, they are closely related. An affinity ceiling is a limit to affinity for a given antigen: there are no mutations that can improve affinity beyond this level. This would result in a truncated response function, undefined beyond the affinity ceiling. A fitness ceiling, on the other hand, is an upper asymptote on the response function. Such a ceiling would result in a limit on affinity for a germinal center reaction, since once cells are well into the upper asymptote of fitness they are no longer subject to selective pressure.”

(2) Lines 566-569: I would like to see this caveat fleshed out more and perhaps mentioned earlier in the paper. While relative affinity is far more important, it is not at all clear to me that absolute affinity can be totally ignored in modeling GC behavior.

This is a great point, we've added a mention of this where we introduce the replay experiment in the Methods:

“It is important to note that this is a much lower level than typical BCR repertoires, which average roughly 5-10% nucleotide shm.”

And expanded on the explanation in the Discussion:

“Some aspects of behavior in the low-shm/early times regime of the extracted GC data are also potentially different to those at the higher shm levels and longer times found in typical repertoires. This is especially relevant to affinity or fitness ceilings, to which we likely have little sensitivity with the current data.”

(3) One other limitation that is worth mentioning, though beyond the scope of the current work to fully address: the evolution of the repertoire is also strongly shaped by competition from circulating antibodies. (Eg: <http://www.ncbi.nlm.nih.gov/pmc/articles/PMC3600904/>, <http://www.sciencedirect.com/science/article/pii/S1931312820303978>). This is irrelevant for the replay experiment modeled here, but still an important factor in general repertoires.

Yes good point, we've added these citations in a new paragraph on between-lineage competition:

“We also neglect competition among lineages stemming from different rearrangement events (different clonal families), instead assuming that each GC is seeded with instances of only a single naive sequence, and that neither cells nor antibodies migrate between different GCs. More realistically for the polyclonal GC case, we would allow lineages stemming from different naive sequences to compete with each other both within and between GCs (Zhang et al. 2013; McNamara et al. 2020; Barbulescu et al. 2025). Implementing competition among several clonal families within a single GC would be conceptually simple and computationally practical in our current software framework. Competition among many GCs, however, would be computationally prohibitive because our time required is primarily determined by the total population size, since at each step we must iterate over every node and every event type in order to find the shortest waiting time. For the monoclonal replay experiment specifically, however, all naive sequences are the same and so the current modeling framework is sufficient.”

Recommendations for the authors:

Reviewing Editor Comments:

The authors are encouraged to follow the suggestions of manuscript re-organization by Reviewer 1, in order to improve readability. We would also like to suggest improving the discussion of the traveling wave model to explain it in a more self-contained way. In

passing, please clarify what is meant by 'steady-state' in that model. A superficial understanding would suggest that the only steady state in that model would be a homogeneous population of antibodies with maximum affinity/fitness.

These are great suggestions. We have substantially rearranged the text according to Reviewer 1's suggestions, especially the Methods, and expanded on and rearranged the traveling wave discussion. We've also clarified throughout that the traveling wave model is assuming steady state with respect to population. In the public response to reviewer 1 above we describe these changes in more detail.

Reviewer #1 (Recommendations for the authors):

I suggest that the organization of the paper be reconsidered. The current methods section is long and at times repetitive, making it impossible to parse in a single reading. Moving some technical details from the main text to an appendix could improve readability. Despite the length of the methods section, many important points, such as justification of choices in model specification or values of parameters, are treated only briefly.

We have rearranged the methods section, particularly the discussion of our model, and have more clearly justified choices of parameter values as described in the public response.

Discussion of similarities and differences with reference to Dewitt et al. 2025 should be revised, as it's currently unclear whether the method presented here has any advantages.

We have expanded this comparison, and emphasized the main disadvantage of the traveling wave approach: there is no way of knowing whether by abstracting away so much biological detail it misses important effects. We have also emphasized that the two approaches use different types of data (time series vs endpoint) which are typically not simultaneously available:

“The clear advantage of the traveling wave model is its simplicity: if its high level view is accurate enough to effectively model the relevant GC dynamics, it is far more tractable. But reproducing low-level biological detail, and making high-dimensional real data comparisons (e.g. Figure 5) to iteratively improve model fidelity, are also useful, providing direct evidence that we are correctly modeling the underlying biological processes. The two approaches also utilize different types of data: we use a single time point, and thus must reconstruct evolutionary history; whereas the traveling wave requires a series of timepoints. The availability of both types of data is a unique feature of the replay experiment, and provides us with the opportunity to directly compare the approaches.”

The results obtained from the same data should be directly compared (can the response function be directly compared to the result in Figure S6D in Dewitt et al., 2025? If yes, it should be re-plotted here and compared/superimposed with Figures 6 and 7). The text mentions the results differ, but it remains ambiguous whether the differences are significant and what their implications are.

We've added a new Figure 8, comparing a modified version of the traveling wave Fig S6D to a new plot derived from our results using the data mimic parameters. While the two plots represent fundamentally different quantities, they do put the results of the two methods on an approximately equal footing and we see nice concordance between them in regions with significant data (they disagree substantially for larger negative affinities). We have also added emphasis to the point that the traveling wave model uses an entirely separate dataset to what we use here.

Other comments:

| (1) l. 80: "[in] around 10 days"?

Text rearranged so this phrase no longer appears.

| (2) l. 96: "an intrinsic rate [given by?] the response function above".

Text rearranged so this phrase no longer appears.

| (3) Figure 1: The. "specific model" could part be expanded and improved to help make sense of model parameters and the order of different processes in the population model. Example values of parameters can be plotted rather than loosely described, (e.g., $y_h + y_c$, the upper asymptotes can be plotted in place of the "yscale determines upper asymptotes" label.

Great suggestion, we've changed the labels.

| (4) The cartoons in the other parts are somewhat cryptic or illegible due to small sizes.

We have added text in the caption linking to the figures that are, in the figure, intended to be in schematic form only.

"Plots from elsewhere in the manuscript are rendered in schematic form: those in "infer on data" refer to Figure 4-figure supplement 1, and those in "simulate with inferred parameters" to Figure 5.

| (5) L. 137: It's not helpful to give numerical values before the definition of affinity. (and these numbers are repeated later).

Good point, we've moved the affinity definition to the previous section, and remove the duplicate range information.

| (6): Table 1: A number of notations are unclear, such as "#seqs/GC" or "mutability multiplier". The double notation for crucial parameters doesn't help. At the moment the table is introduced, the columns make little sense to the reader, and it's not well specified what dictates the choice or changes of parameter values or ranges.

We've moved the table further down until after the parameters have been introduced, and clarified the indicated names.

| (7) l. 147: Choices of model are not justified and appear arbitrary (e.g., why death events happen at one of two rate).

We have clarified the reasoning behind having two death rates.

| (8) l.151: "happened on the edges of developing phylogenetic tree" - ambiguous: do they accumulate at cell divisions? What is a "developing tree"?

We have removed this ambiguous phrasing.

| (9) l.161: This paragraph is particularly dense.

We have rearranged this section of the methods, and split up this paragraph.

| (10) l. 164: All the different response functions for different event types? Or only the one for birth, as stated before?

Yes. This has been clarified.

| (11) l.167: Does the statement in the bracket refer to a unit?

This has been clarified.

| (12) l. 169: Discussion of the implementation seems too detailed.

Hopefully the rearranged description is clearer, but we worry that removing the details of events selection would leave some readers confused.

| (13) l. 186: Why describe the methods that, in the end, were not used? Similarly, as a mention of "variety of response functions" seems out of place if only one choice is used throughout the paper. eq. (2): that's $m\{-1\}$ from eq. (1). Having the two equations using the same notation is confusing.

We've moved the mention of alternatives to the Discussion, where it is an important source of uncontrolled systematic uncertainty, and removed the extra equation.

| (14) l. 206: Unclear what "thus" refers to.

Removed.

| (15) l.211: What does "neglecting y_h " mean?

This has been clarified.

| (16) l. 242: Unclear what "this" refers to.

Clarified.

| (17) l. 261: What does "model independence" refer to in this context?

From the sigmoid model. Clarified.

| (18) l. 306: What values for which parameters? References?

We have clarified and updated this statement - it was out of date, corresponding to the analysis before we started fitting non-sigmoid parameters.

"In addition to the four sigmoid parameters, which we infer directly, there are other parameters in Table 1 about which we have incomplete information. The carrying capacity method and the choice of sigmoid for the response function represent fundamental model assumptions. We also fix the death rate for nonfunctional (stop) sequences, which would be very difficult to infer with the present experiment. For others, we know precise values from the replay experiment for each GC (time to sampling, # sampled cells/GC), but use a somewhat wider range for the sake of generalizability. The mutability multiplier is a heuristic factor used to match the SHM distributions to data. The naive birth rate is determined by the sigmoid parameters, but has its own range in order to facilitate efficient simulation.

For two of the three remaining parameters (carrying capacity and initial population), we can ostensibly choose values based on the replay experiment. These values carry significant uncertainty, however, partly due to inherent experimental uncertainty, but also because they may represent different biological quantities to those in simulation. For instance, an experimental measurement of the number of B cells in a germinal center might appear to correspond closely to simulation carrying capacity. However if germinal centers are not well mixed, such that competition occurs only among nearby cells, the "effective" carrying capacity that each cell experiences could be much smaller.

Fortunately, in addition to the neural network inference of sigmoid parameters, we have another source of information that we can use to infer non-sigmoid parameters: summary statistic distributions. We can use the matching of these distributions to effectively fit values

for these additional unknown parameters. We also include the final parameter, the functional death rate, in these non-sigmoid inferred parameters, although it is unconstrained by the replay experiment, and it is unclear whether it is uniquely identifiable.”

| (19) l. 326: *"is interpreted as having" or "corresponds to"?*

Changed.

| (20) l. 340: *Not sure what "encompassing" means in this context.*

Clarified.

| (21) l. 341: *"We do this..." -- I think this sentence is not grammatical.*

Fixed.

| (22) l. 348: *"on simulation" -- "from simulated data"?*

Indeed.

| (23) l. 351: *"top rows", the figures only have one row.*

Fixed.

| (24) Figure 2: *It's difficult to tell from the loss function itself whether inference on simulated data works well. Why not report the simulated and inferred response functions? The equivalent plots in Figure 5 would also be informative. Has inference been tested for different "sigmoid parameters" values?*

This is an important point that was not clear, thanks for bringing it up. We have expanded on and emphasized the differences between these samples and the reasoning behind their different evaluation choices. Briefly, we can't display true vs inferred response functions on the training samples since the curves for each GC are different -- the plot would be entirely filled in with very different response function shapes. This is why we do actual performance evaluation on the "data mimic" samples, where all GCs have the same parameters. Summary stats (like Fig 5) for the training sample are in Fig 5 Supplement 2.

| (25) l. 354: *Unclear what "this" refers to.*

Removed.

| (26) l. 355: *We assume the parameters are the same?*

Yes, we assume all data GCs have the same parameters. We have added emphasis of this point.

| (27) Figure 4: *Is " λ " the fitness? Should be typeset as λ ?*

Our convention is to add the subscript when evaluating fitness on individual cells, but to omit it, as here, when plotting the response function as a whole.

| (28) l. 412: *"[a] carrying capacity constraint".*

Fixed.

Reviewer #2 (Recommendations for the authors):

(1) In 2 places, you state that observed affinity ranged from -37 to 3, but I assume that the lower bound should be -3.7.

The -37 was actually correct, but we had mistakenly missed updating it when we switched to the latest (current) version of the affinity model. We have updated the values, although these don't really have any effect on the model since we only infer within bounds in which we have a lot of points:

“Affinity is $\emptyset$ for the initial unmutated sequence, and ranges from -12.2 to 3.5 in observed sequences, with a mean median of -0.3 (0.3).

(2). I had to look up the Vols nicker paper to understand the tree encoding: It would be nice to spend another sentence or two on it here for those who aren't familiar.

Great point, we have added the following:

“We encode each tree with an approach similar to Lambert et al. (2023) and Thompson et al. (2024), most closely following the compact bijective ladderized vector (CBLV) approach from Voznica et al. (2022). The CBLV method first ladderizes the tree by rotating each subtree such that, roughly speaking, longer branches end up toward the left. This does not modify the tree, but rather allows iteration over nodes in a defined, repeatable way, called inorder iteration. To generate the matrix, we traverse the ladderized tree in order, calculating a distance to associate with each node. For internal nodes, this is the distance to root, whereas for leaf nodes it is the distance to the most-recently-visited internal node (Voznica et al., 2022, Fig. 2). Distances corresponding to leaf nodes are arranged in the first row of the matrix, while those from internal nodes form the second row.”

(3) On line 351, you refer to the "top rows of Figure 2 and Figure 3," but each only has one row in the current version. I think it should now be "left panel."

Fixed.

(4) How many vertical dashed lines are in the left panel of the bottom row of Figure 7? I think it's more than one, but can't tell if it is two or three...

Nice catch! There were actually three. We've shortened them and added a white outline to clarify overlapping lines.

(5) Would the model be applicable to GCs with multiple naive founders of different affinities? Or would more/different parameters be needed to account for that?

The model would be applicable, but since the time required for our simulation scales roughly with the total simulated population size, we could probably only handle competition among at most a couple of GCs. Some sort of "migration strength" parameter would be required for competition among GCs (or within one GC if we don't want to assume it's well-mixed), but that doesn't seem a terrible impediment. We've added the following:

“We also neglect competition among lineages stemming from different rearrangement events (different clonal families), instead assuming that each GC is seeded with instances of only a single naive sequence, and that neither cells nor antibodies migrate between different GCs. More realistically for the polyclonal GC case, we would allow lineages stemming from different naive sequences to compete with each other both within and between GCs (Zhang et al. 2013; McNamara et al. 2020; Barbulescu et al. 2025). Implementing competition among several clonal families within a single GC would be conceptually simple and computationally practical in our current software framework. Competition among many GCs, however, would be computationally prohibitive because our time required is primarily determined by the total population size, since at each step we must iterate over every node and every event type in order to find the shortest waiting time. For the monoclonal replay experiment specifically, however, all naive sequences are the same and so the current modeling framework is sufficient.”

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